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APPROVAL SHEET FORMAT

This Group Project entitled “Smart Internship Placement for University Students” has been read and approved as meeting the preliminary project requirements of the Department of Computer Science in partial fulfillment of the requirements for the award of a Bachelor of Science degree in Computer Science, University of Gondar. Gondar, Ethiopia.

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ACRONYMS

- ❖ AI: Artificial Intelligence
- ❖ API: Application Programming Interface
- ❖ CDN: Content Delivery Network
- ❖ CORS: Cross-Origin Resource Sharing
- ❖ CV: Curriculum Vitae
- ❖ ER: Entity-Relationship
- ❖ GPS: Global Positioning System
- ❖ HTTPS: Hypertext Transfer Protocol Secure
- ❖ IDE: Integrated Development Environment
- ❖ JSON: JavaScript Object Notation
- ❖ JWT: JSON Web Token
- ❖ LLM: Large Language Model
- ❖ MERN: MongoDB, Express.js, React.js, Node.js
- ❖ MitM: Man-in-the-Middle (Attack)
- ❖ NLP: Natural Language Processing
- ❖ PM2: Process Manager 2
- ❖ RBAC: Role-Based Access Control
- ❖ SDLC: System Development Life Cycle
- ❖ SSL: Secure Sockets Layer
- ❖ TF-IDF: Term Frequency-Inverse Document Frequency
- ❖ TLS: Transport Layer Security
- ❖ UIL: University-Industry Linkage
- ❖ UI/UX: User Interface / User Experience
- ❖ UML: Unified Modeling Language
- ❖ VSM: Vector Space Model
- ❖ XSS: Cross-Site Scripting

EXECUTIVE SUMMARY

The Smart Internship Placement for University Students is a data-driven system designed to automate and optimize the transition from academic learning to industrial practice. Currently, the internship placement process at the University of Gondar relies on manual, paper-based workflows, often resulting in a systemic mismatch between student competencies and industry requirements. This project addresses these challenges by introducing a "Managed Brokerage" framework where the University-Industry Linkage (UIL) Directorate and Department Coordinators maintain institutional oversight through a verified digital environment.

The primary innovation of this system is its Intelligent Matching Engine. To move beyond high-level descriptions, the platform implements Natural Language Processing (NLP) using the Vector Space Model (VSM). Specifically, the system utilizes TF-IDF (Term Frequency-Inverse Document Frequency) vectorization paired with the Cosine Similarity algorithm. This allows the system to compute a precise mathematical similarity score between a student's academic profile, This ensures that placements are based on technical compatibility rather than random assignment.

Developed using the MERN stack (MongoDB, Express.js, React.js, and Node.js), the platform consists of three integrated portals: a Student Portal for persistent profile management, an , and a Centralized Admin Panel for institutional monitoring. The significance of this project lies in its ability to reduce administrative overhead for the UIL office while providing students with meaningful professional exposure aligned with their curriculum. In conclusion, this system provides a robust, scalable, and intelligent linkage between academia and the industry, enhancing the overall quality of professional preparation at the University of Gondar.

CHAPTER ONE

INTRODUCTION

The integration of digital technologies into higher education plays a pivotal role in bridging the gap between academic learning and professional industry practice. In the modern educational landscape, the process of securing internships is shifting from traditional manual methods to intelligent, data-driven platforms. However, at the University of Gondar, this transition faces significant challenges. The current internship placement process relies heavily on manual coordination, paper-based workflows, and physical filing systems. This often leads to inefficiencies in which students are randomly assigned to organizations, without considering their specific technical skills or field of study.

To address these challenges, this project introduces the "Smart Internship Placement for University Students." This is an AI-Powered Managed System designed to automate, streamline, and professionally manage the student-industry linkage process. Unlike open job boards, this platform operates on a "Managed Brokerage" model, ensuring that every internship placement is verified, curriculum-aligned, and monitored for quality.

A critical aspect of this system is its integration with the University of Gondar's administrative hierarchy. Addressing the question of institutional responsibility within the university's structure—which spans various Campuses, Institutes, Colleges, and Schools—the system defines clear operational owners. The University-Industry Linkage (UIL) Directorate serves as the central authority, responsible for verifying the legitimacy of host organizations and setting institutional standards. At the operational level, the responsibility is delegated to Departmental Coordinators (specifically within units like the Department of Computer Science). These coordinators act as the primary system administrators who audit student profiles, validate academic eligibility, and approve the final forwarding of candidates to industry partners.

Technologically, the system eliminates the subjectivity of manual selection by utilizing an Intelligent Matching Engine. By leveraging Natural Language Processing (NLP) and the Vector Space Model (VSM), the platform mathematically analyzes the compatibility between a student's academic profile and an internship's technical requirements. This ensures that placements are based on merit and skill relevance.

1.1. BACKGROUND OF THE PROJECT

The facilitation of student internship opportunities has historically relied on manual and uncoordinated methods and techniques that struggle to address the professional complexities of the modern digital job market.

Despite significant advancements in recruitment technology, a substantial gap remains in comprehensive platforms capable of managing university-specific linkages, particularly those that seamlessly integrate intelligent skill-based matching and robust administrative oversight. The transition from an academic environment to industrial practice requires more than just a job board; it requires a managed ecosystem that ensures students are placed in roles aligned with their fields of study.

The genesis of the Smart Internship Placement for University Students was rooted in observations of numerous students struggling to navigate the transition from academia to industry. These firsthand encounters underscored the need for a platform that not only simplifies the search for opportunities but also creates a verified, high-trust professional ecosystem. Current career placement methods tend to offer only random assignments or unorganized listings, whereas our system is crafted to address the dynamic and multifaceted requirements of modern university students and industry partners through a coordinated institutional framework.

While general job boards exist globally, valuable insights were gleaned from the logic of managed professional recruitment agencies during the design phase. Notably, the managed brokerage feature—where University-Industry Linkage (UIL) coordinators and Departmental staff act as central intermediaries—was inspired by the screening protocols of specialized human resource firms. This concept has been adapted to ensure that students remain focused on roles that match their technical expertise and academic curriculum, effectively transforming a fragmented search process into a controlled and results-oriented career pipeline.

On the technical front, our technology choices were driven by the imperative for scalability and intelligent data processing. We selected the MERN stack (MongoDB, Express, React, and Node.js) for its ability to build responsive dashboards and handle persistent professional profiles. Coupled with Natural Language Processing (NLP) libraries such as 'Natural' for Node.js and 'Compromise' for text tokenization, this technological stack provides an efficient framework for text-based

similarity analysis—a crucial component in maintaining the smart matching nature of the platform. These tools allow the implementation of the Vector Space Model (VSM) to ensure high-precision matching.

In summary, the Smart Internship Placement for University Students is a deliberate response to the shortcomings of manual and random student placements. By drawing on industry recruitment standards and incorporating proven AI concepts for data matching, our solution delivers a robust, user-centric platform that enhances both the professional readiness and industry engagement of university students.

1.2 STATEMENT OF THE PROBLEM AND JUSTIFICATION

The current internship placement management at the University of Gondar, specifically within the Department of Computer Science, operates on a highly fragmented, manual, and paper-intensive workflow that creates a significant administrative bottleneck for the Department Head and academic staff. The process begins with a cumbersome logistical challenge where students must physically queue at the department office to request generic support letters. The Department Head is frequently overwhelmed with the task of manually signing and stamping hundreds of these letters without having a centralized mechanism to verify if the organizations the students are approaching are legitimate or capable of providing curriculum-aligned training. This lack of data-driven oversight forces the Department Head to authorize placements based on availability rather than technical compatibility, leading to a systemic mismatch where Computer Science students are often accepted by organizations that assign them to non-technical, clerical roles such as data entry or photocopying, which effectively nullifies the academic value of the internship program.

Furthermore, the reliance on physical documentation creates a chaotic feedback loop between the student and the department. Once a student manages to find an organization, they must physically transport the "Acceptance Letter" back to the department office to confirm their placement. These critical documents are typically stored in physical folders or disorganized filing cabinets, making it nearly impossible for the Department Head to quickly retrieve student records, verify placement history, or generate real-time reports. This manual data handling poses a high risk of document loss and data redundancy, where the department loses track of which student is placed where. Consequently, the Department Head lacks the necessary visibility to monitor the students' progress effectively. Since there is no digital link between the department and the industry supervisors, the

university remains in the dark regarding the student's daily attendance and performance until the very end of the semester when physical logbooks are submitted, by which time it is too late to intervene if a student has been facing technical challenges or exploitation.

Justification of the Project

The development of the Smart Internship Placement System is justified by the urgent need to resolve the systemic inefficiencies described above. As the student population grows, maintaining the status quo of manual coordination is no longer sustainable.

Primarily, this project is necessary to **ensure academic quality**. By shifting from random assignment to a system that verifies the correlation between student skills and industry requirements, the university can guarantee that internships serve their intended educational purpose.

Secondly, the project is justified by the need for **administrative modernization**. Digitizing the workflow will drastically reduce the workload on department coordinators, eliminate data loss, and provide the university with analytics on industry partnerships.

Finally, the system is essential for **professional accountability**. Establishing a verified digital environment protects students from unverified organizations and ensures that industry partners receive qualified candidates, thereby strengthening the long-term relationship between the University of Gondar and the industrial sector.

1.3 OBJECTIVE OF THE PROJECT

1.3.1 General objective

The general objective of this project is to design and develop a Smart Internship Placement System for University Students that transforms the current manual career linkage process into a secure, intelligent, and professionally managed digital environment.

1.3.2 Specific objectives

To achieve the general goal of the project, the following specific objectives are defined:

- ❖ To analyze the existing manual internship placement workflow and identify specific operational bottlenecks and data gaps.

- ❖ To design and develop a secure Role-Based Access Control (RBAC) architecture that handles distinct authentication for Students, Industry Partners, and UIL Coordinators (Department Head).
- ❖ To develop a Persistent Academic Profile module that allows students to build verifiable digital CVs and input technical skills for AI processing.
- ❖ To create a Vacancy Management Console for verified Industry Partners to post curriculum aligned internship opportunities.
- ❖ To implement an AI-Driven Smart Matching Engine using TF-IDF and Cosine Similarity algorithms to mathematically rank student compatibility with internship requirements.
- ❖ To engineer a Managed Brokerage Dashboard that empowers UIL Coordinators (Department Head) to audit match scores and manually forward qualified profiles to employers.
- ❖ To develop a Real-Time Notification System to track application status updates and facilitate a feedback loop for hiring decisions.

Integration of Modules.

The modules of the Smart Internship Placement for University Students are engineered to operate synchronously, creating a unified and efficient institutional framework for professional linkage.

The Matching and Communication Module serves as the core infrastructure for real-time interaction, enabling University-Industry Linkage (Department Head) coordinators, students, and industry partners to exchange application status updates and collaborate during the internship placement lifecycle. It integrates seamlessly with the Persistent Profile Module, which ensures that coordinators have immediate access to comprehensive student data, including verified technical competencies, academic project history, and course specializations, directly within the platform. Together, these modules enhance the linkage experience by fostering transparency and ensuring that students are placed in environments that maximize their academic potential.

The Administration and Brokerage Control Module acts as the central hub for managing the system, allowing UIL Directorate (Department Head) to oversee host organization registrations, verify internship role legitimacy, and manage system-wide workflows. It works in close coordination with the Role-Based Authentication Module, which enforces strict access controls based on institutional roles. This integration ensures that only authorized academic staff can access sensitive professional

records or perform administrative actions such as auditing and forwarding student profiles to industry partners. This ensures the system remains secure, well-organized, and compliant with university data protection policies.

The AI Smart Matching Module plays a critical role in maintaining the quality and objectivity of the placement process. By utilizing Natural Language Processing (NLP) - specifically the Vector Space Model (VSM) through TF-IDF and Cosine Similarity - the module ranks student applicants based on the mathematical alignment between their academic profiles and the specific technical requirements of an internship role. It integrates with the Role-Based Authentication Module to enforce professional restrictions and with the Opportunity Finder Module to allow students to discover relevant roles within the managed environment. This ensures that the linkage process remains productive while strictly adhering to the quality standards set by the university administration.

The Data Security and Privacy Module ensures that the system operates with high integrity, protecting sensitive student and institutional information from unauthorized access. It integrates with the Role-Based Authentication Module and the Brokerage Control Module to enforce rigorous data-handling policies, creating a secure digital environment for all stakeholders. Additionally, the Web-Based Admin Panel Module provides coordinators with a centralized interface for auditing placements, tracking student progress, and monitoring system analytics, integrating with the core management modules to ensure smooth and secure institutional operations.

Finally, the User Dashboard Module enhances the professional experience by allowing students and industry partners to manage their respective interactions without disrupting the managed brokerage workflow. It integrates with the AI Smart Matching Module to ensure that information is filtered and prioritized based on academic relevance, maintaining the integrity and efficiency of the overall university-industry linkage environment.

1.4 SCOPE OF THE PROJECT

The Smart Internship Placement System is designed as a web-based platform to automate and streamline the internship coordination process at the University of Gondar. The primary scope of the project focuses on developing three integrated portals: a Student Dashboard for profile management and opportunity discovery, an Employer Interface for posting internship vacancies, and a Centralized Admin Panel for UIL coordinators (Department Head) to oversee the entire linkage process. The system enables students to build a digital professional profile by inputting their academic history, technical skills, and project portfolios. These profiles are then processed by an intelligent matching engine that automatically compares student competencies with industry requirements to suggest the most relevant placement opportunities. Furthermore, the system incorporates a "Managed Brokerage" workflow, which ensures that all student applications and industry internship postings are audited and verified by university coordinators before any formal connection is established.

However, to ensure the project remains feasible within the given timeframe, certain functionalities are explicitly excluded from the current scope. The system will operate as a standalone platform and will not integrate directly with the university's main registrar database due to data privacy and security constraints; therefore, student grades and academic records must be entered manually or uploaded by the students. Additionally, the project does not include features for real-time monitoring such as GPS tracking of students at their workplace, nor does it support financial transaction processing for stipend payments. The system is also limited to a responsive web application and does not include the development of native mobile applications (Android/iOS) or automated resume parsing capabilities from PDF documents.

1.5 LIMITATIONS OF THE PROJECT

While the Smart Internship Placement System offers significant improvements over the manual workflow, it operates within specific boundaries and constraints. The following points outline the limitations of the current project:

1. Domain Specificity (Departmental Constraint):

The system's AI matching logic (NLP) is currently optimized specifically for

the Department of Computer Science and Information Technology. The keyword extraction and similarity algorithms are tuned to recognize technical terminologies (e.g., "React," "Network Security," "Database"). Consequently, the system is not immediately scalable to other departments like Medicine, Law, or Social Sciences without significant reconfiguration of the matching dataset and logic.

2. Linguistic Constraints (English Only):

The system supports only the English language for both user interfaces and data processing. It cannot process internship job descriptions posted in Amharic or other local languages. If an industry partner submits a vacancy in a local language, the NLP engine will fail to extract keywords, rendering the matching score inaccurate.

3. Lack of Offline Functionality:

The platform is entirely web-based and requires a continuous internet connection to function. It does not possess offline synchronization capabilities. In scenarios where there is an internet outage at the university or within the host organizations, users cannot access their dashboards, view statuses, or upload documents until connectivity is restored.

4. Reliance on Self-Reported Proficiency:

While the system verifies that a student has listed a skill, it cannot algorithmically verify the depth or truthfulness of that skill. For instance, if a student claims to be an "Expert" in Python, the system calculates the match based on that claim. It does not include an integrated technical assessment module (like a coding test) to validate the actual competency level of the student before the interview stage.

1.6 SYSTEM DEVELOPMENT METHODOLOGY

To ensure the successful delivery of the Smart Internship Placement System, the **Hybrid Iterative and Incremental Model** has been selected. This approach was chosen not merely as a standard procedure, but because it directly aligns with the system's architectural division and the complexity of its matching logic.

1.6.1 System Development Approach

Chosen Methodologies: Hybrid Iterative and Incremental Model

Why the Incremental Approach? (Modular Development)

The system is composed of three distinct functional environments: the Student Dashboard, the Employer Portal, and the Coordinator (Admin) Panel. The Incremental approach allows the development team to build and deploy these modules independently.

- **Increment 1:** We will first develop the **Student Profile Module** to ensure data collection capability.
- **Increment 2:** We will then build the **Employer Module** to allow vacancy posting.
- **Increment 3:** Finally, we will integrate the **Brokerage/Admin Module** to link the two. This ensures that at the end of each increment, a functional part of the system is ready for testing, reducing the risk of total system failure at the end of the year.

Why the Iterative Approach? (Logic Refinement)

The core value of this project lies in the "Smart Matching" logic, which utilizes an algorithm to compare student skills with job requirements. Unlike standard CRUD (Create, Read, Update, Delete) operations, this logic requires repeated tuning.

- In the first iteration, the matching might only look for exact keyword matches.
- In subsequent iterations, we will refine the algorithm to handle synonyms and skill weighting.

The Iterative model allows us to revisit and improve the accuracy of the matching engine repeatedly based on test results, without disrupting the already finished modules.

Conclusion

By combining these two approaches, the team can deliver the project components part-by-part (Incremental) while continuously refining the complex brokerage and matching rules (Iterative) to meet the department's quality standards.

1.6.2 System Development Tools

To build the Smart Internship Placement for University Students, a combination of high-performance tools and modern web technologies was utilized. These tools were selected to ensure the system's scalability, security, and the effective implementation of the AI-driven matching engine.

Table 1.1: System Development Tools and Technologies

Category	Tool/ Technology	Purpose and Application
Documentation	Microsoft Word	To facilitate collaborative authoring and version

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		control of the project documentation and technical reports.
System Design	Draw.io	To model complex system architectures, including Use Case diagrams, ER diagrams, and managed brokerage sequence diagrams.
Frontend	React.js	Chosen for its component-based architecture to develop dynamic, responsive academic dashboards for students and coordinators.
Backend	Node.js & Express.js	To engineer a robust server-side environment capable of managing complex API services and the core brokerage logic.
Database	MongoDB	A NoSQL database was selected for its flexibility in storing semi-structured persistent student profiles and internship role data.
AI/NLP Engine	Natural (Node.js Library)	Utilized to implement the NLP pipeline, specifically for TF-IDF vectorization and Cosine Similarity scoring.
Version Control	GitHub	Employed for collaborative source code management, branch tracking, and maintaining the project's development history.
UI/UX Design	Figma	To create high-fidelity prototypes and wireframes for the Student and Administrative portals before actual development.

By leveraging the Iterative and Incremental Model alongside the advanced technology stack mentioned above, the development process remains dynamic, user-centric, and highly adaptable to the evolving professional requirements of the University of Gondar's internship linkage environment.

1.7 SIGNIFICANCE OF THE PROJECT

The implementation of the Smart Internship Placement System holds substantial value for the University of Gondar, the students, and the industrial partners. Its significance is primarily observed in the transition from a disorganized manual process to a centralized, data-driven management system. Specifically, the project contributes in the following key areas:

Enhancement of Academic and Professional Alignment:

The most critical contribution of this system is the elimination of random placement. By utilizing the intelligent matching engine, the system ensures that students are assigned to internships that directly correlate with their specific technical skills and field of study. This significance lies in maximizing the educational value of the internship program, ensuring that Computer Science students are placed in roles where they can apply their coding and system analysis skills, rather than engaging in unrelated clerical work.

Operational Efficiency and Resource Optimization:

For the University-Industry Linkage (UIL) office, the project significantly reduces the administrative burden. By digitizing the workflow, the system eliminates the time-consuming tasks of manual file sorting, physical application delivery, and face-to-face status inquiries. This allows academic coordinators to focus on strategic partnership building rather than clerical maintenance, thereby streamlining the entire academic calendar's internship phase.

Institutional Oversight and Data Integrity:

The system establishes a verified digital repository for student records and industry agreements. Its significance is evident in its ability to provide the university with real-time oversight regarding where students are placed and the legitimacy of the host organizations. This creates a secure and transparent environment that prevents fraudulent internships and ensures that the university maintains a persistent history of its industry linkages for future accreditation and auditing purposes.

1.8 BENEFICIARIES OF THE PROJECT

The Smart Internship Placement for University Students provides multi-dimensional benefits to various stakeholders within the educational and professional community. The primary beneficiaries and the specific advantages they gain are as follows:

1. University-Industry Linkage (Department Heads):

Academic administrators, specifically Department Heads, benefit significantly from the centralized oversight provided by the platform. The system empowers them to efficiently manage the placement of large student batches without the administrative burden of manual sorting. Department Heads gain visibility into whether the internships offered by companies actually meet the curriculum standards of the department. This ensures that the academic integrity of the internship program is maintained and that placements are approved based on educational value rather than logistical convenience.

2. Students:

Students experience a more transparent and accessible process for securing industrial placement. The core benefit is the ability to discover internship roles that align **closely** with their academic background and current skill level. Instead of being assigned randomly to irrelevant sectors, students are matched with organizations where they can apply their classroom knowledge practically. Additionally, the system provides them with real-time status updates on their application, eliminating the anxiety and uncertainty caused by the traditional lack of communication.

3. Industry Partners (Host Organizations):

Host organizations benefit from a streamlined process of accepting interns who are academically prepared for the tasks at hand. Unlike the recruitment of full-time employees, the goal here is to minimize the **training burden** on industry supervisors. By receiving students whose skills have been pre-matched to the internship requirements, companies can spend less time on basic instruction and more time mentoring students on productive tasks. This transforms the internship from a supervisory burden into a mutually beneficial engagement where students contribute fresh perspectives to the organization.

1.9 FEASIBILITY STUDY

The feasibility study evaluates the practicality and viability of implementing the Smart Internship Placement for University Students, ensuring that it aligns with technical, operational, legal, and scheduling requirements. The project aims to modernize the student career linkage process by enabling coordinators to facilitate AI-driven placements in a controlled digital environment. This study analyzes whether the proposed system can be effectively developed, integrated into the university structure, and maintained within the existing constraints.

1. Technical feasibility

The technical feasibility assessment confirms that the proposed "Smart Internship Placement System" can be successfully developed, deployed, and maintained using currently available resources and expertise. The justification for the project's technical viability is based on three core pillars: availability of technology, computational efficiency of the chosen algorithms, and the team's proficiency.

Availability and Maturity of the Technology Stack:

The system is architected using the **MERN Stack (MongoDB, Express.js, React.js, Node.js)**. This choice ensures technical feasibility because these technologies are open-source, widely supported, and highly compatible with web-based institutional platforms. MongoDB provides the necessary schema flexibility to store diverse student profiles, while Node.js offers the asynchronous processing power required to handle concurrent requests from students and coordinators. Since these tools are free and platform-independent, there are no barriers related to licensing or proprietary hardware requirements.

Feasibility of the AI/NLP Component:

A critical technical challenge is the implementation of the "Smart Matching" engine. To ensure this is computationally feasible within a university server environment, the system avoids resource-heavy Large Language Models (LLMs). Instead, it utilizes the **Vector Space Model (VSM)** with **TF-IDF vectorization** and **Cosine Similarity**. This approach is mathematically lightweight, allowing the matching algorithm to run efficiently on standard hardware without requiring expensive GPUs or cloud infrastructure. Libraries such as 'Natural' for Node.js are mature and readily available to implement this logic effectively.

Team Competence and Development Tools:

The project team possesses the requisite skills in full-stack JavaScript development and algorithmic design to execute the project. The development environment relies on standard, accessible tools

such as **Visual Studio Code** for coding, **Postman** for API testing, and **Git** for version control. Given that the hardware requirements (standard laptops and local servers) are already available to the team, and the software tools are open-source, there are no technical bottlenecks preventing the successful completion of the project.

2. Operational Feasibility

Operational feasibility evaluates whether the Smart Internship Placement for University Students will function effectively within the real-life organizational structure of the university and if it meets the practical needs of its users.

- ✓ **Real-Life Operation and Workflow Integration:** The system is engineered to digitize and optimize the existing manual internship placement workflow. It mirrors the actual institutional process where the University-Industry Linkage (UIL) Office and Department Coordinators act as the central authority. By automating the candidate-to-organization matching, the system transitions the current "random" placement method into a structured, data-driven operation that works seamlessly within the university's academic calendar.
- ✓ **Organizational Structure Alignment:** The platform respects the existing hierarchy of the University of Gondar. It provides dedicated access levels for UIL Directors at the institutional level or Coordinators at the department level, ensuring that the digital tool supports, rather than disrupts, the university's administrative structure.
- ✓ **User Acceptance and Adaptability:** The system is designed with a high focus on usability and intuitive UI/UX. Preliminary analysis suggests high user acceptance because the platform directly solves the "pain points" of coordinators (manual file sorting) and students (lack of placement visibility). By providing a transparent and efficient environment, the system gains immediate buy-in from all stakeholders.
- ✓ **Staff Skills and Training Requirements:** The university's administrative staff already possesses the necessary digital literacy to operate web-based platforms. The system's interface is built following modern standards (Material UI), making it accessible without the need for advanced technical training. A brief orientation session and a provided user manual will be sufficient to ensure the staff can effectively manage the "Brokerage Logic" and oversee the AI matching scores.

3. Legal Feasibility

The legal feasibility confirms that the system complies with institutional policies, national data protection standards, and professional privacy laws.

- ✓ **User Data Privacy and Compliance:** The system implements rigorous data protection mechanisms to safeguard students' professional records. By utilizing Role-Based Access Control (RBAC) and JWT-based encryption, the platform ensures that sensitive data, such as student contact information and academic project details, is protected from unauthorized access. This aligns with global and national data privacy standards, ensuring that personal information is only visible to verified academic staff and approved industry partners.
- ✓ **University and Industry Regulations:** The system's operational logic is fully aligned with the University of Gondar's Industry Linkage and Student Placement Policies. It ensures that the internship roles posted by organizations meet the academic standards required by the university before a student is placed. This guarantees that the linkage remains a curriculum-based educational experience.
- ✓ **Civil Service and Employment Standards:** Per the teacher's guidance, the system is designed to distinguish academic internships from permanent civil service recruitment. It complies with employment standards by focusing strictly **on temporary educational placements**. The inclusion of stipend transparency ensures that the system promotes fair treatment of students while adhering to national guidelines regarding intern compensation and workplace safety.
- ✓ **Software Licensing and Intellectual Property:** The platform is developed using open-source technologies (MERN Stack), which avoids proprietary legal issues and ensures that the university retains full ownership of the system's deployment and data without incurring high licensing costs

4. Economic Feasibility

The economic feasibility, often referred to as cost-benefit analysis, evaluates whether the financial investment required for the development and operation of the Smart Internship Placement for University Students is justified by the benefits it delivers to the institution. For this project, the economic feasibility is confirmed through the utilization of cost-effective development tools and the long-term operational efficiency it provides.

A. Development Costs

The development phase of this project is highly cost-effective due to the following factors:

- ✓ **Open-Source Ecosystem:** The system is built using the MERN stack, where all core technologies (MongoDB, Express, React, Node.js) and AI libraries are open-source and free of licensing fees. This significantly reduces the initial capital expenditure.
- ✓ **Human Resource Utilization:** As a final year academic project, the labor costs are neutralized since the development is conducted by the project team members as part of their academic requirements.
- ✓ **Hardware and Software Tools:** The development utilizes existing personal hardware (laptops) and free professional IDEs (VS Code), eliminating the need for additional infrastructure investment during the creation phase.

B. Operational and Maintenance Costs

Once deployed, the system maintains a low financial footprint:

- ✓ **Cloud Hosting Efficiency:** The platform can be hosted on scalable cloud environments (such as AWS free tier, Render, or Vercel) which offer low-cost or no-cost entry points for academic applications.
- ✓ **Resource Optimization:** The automation of the placement cycle reduces the need for extensive manual labor in the University-Industry Linkage (Department Head) office, translating into long-term savings in administrative hours and institutional resources.

C. Benefit Analysis (Return on Investment)

The economic value of the system is realized through qualitative and quantitative gains:

- ✓ **Time Efficiency:** By automating the screening and matching process using NLP and Cosine Similarity, the time required for student placement is reduced from weeks to seconds. This time-saving represents a significant indirect financial benefit for the university staff.

- ✓ **Reduction in Material Costs:** Transitioning from a paper-based manual system to a digital platform eliminates the recurring costs of paper, printing, and physical file storage.
- ✓ **Enhanced Institutional Value:** Improving the accuracy of internship placements increases student employability and strengthens industry partnerships, providing intangible yet high-value economic benefits to the University of Gondar's reputation.

Conclusion: Given the minimal development costs achieved through open-source tools and the substantial long-term benefits in administrative efficiency and resource optimization, the project is economically feasible and provides a high return on investment for the institution.

1.10 PROJECT SCHEDULE

The project schedule defines the strategic timeline for each phase of the Smart Internship Placement for University Students. It serves as a critical project management tool for tracking milestones, ensuring the timely completion of deliverables, and managing academic resources effectively. The following schedule outlines the chronological progression of the project—from the initial requirement gathering phase to the final system implementation and validation.

Phase	Activity Description	Planned Dates
Phase 1	Project Initiation & Planning	Dec 24 – Jan 07, 2026
Phase 2	Requirements Analysis & Documentation	Jan 08 – Jan 21, 2026
Phase 3	System Design & UI/UX Prototyping	Jan 22 – Feb 11, 2026
Phase 4	AI Engine Dev (NLP & Similarity Engine)	Feb 12 – Mar 04, 2026
Phase 5	Full-Stack Development (MERN Portals)	Mar 05 – Apr 15, 2026
Phase 6	System Testing & Accuracy Validation	Apr 16 – May 06, 2026
Phase 7	Final Documentation & Project Defense	May 07 – Jun 10, 2026

Table 1.1: Project Schedule Table



1.11 PROJECT BUDGET

The project budget provides an itemized breakdown of the estimated financial resources required to implement the Smart Internship Placement for University Students successfully. It includes all expected costs, such as hardware, software, materials, and other essential resources. This budget ensures that the project remains within financial constraints while covering all necessary components for development and deployment. The goal is to allocate the budget efficiently, allowing for the smooth execution of the project and ensuring that no critical aspects are overlooked.

NO	Item name	Quantity	Price/Unit(Bir ₹)	Total price
1.	Flash Drive 64GB	1	800	800
2.	Laptop	6	45,000	270,000
Total Price				270,800

Table 1.11 Hardware cost

NO	Item Name	Price
1.	VS Code Editor	Free
2.	MongoDB Compass	Free
3.	Draw.io / Figma	Free

Table 1.12 Software cost

NO	Item name	Quantity	Price/Unit(Bir	Total price
1.	Pen	3	25	75
2.	A4 Paper (Ream)	100	5	500
Total Price				575

Table 1-13 Material Cost

NO	For phone calls and Data	Cost(Birr)
1.	With our team members	100
2.	With our advisor	50
Total Price		150

Table 1-14 Communication Cost

Total Cost=Hardware cost + Software cost + Material Cost + Communication Cost

$$= 270,800 + 0.0 + 575 + 150$$

$$= \mathbf{271,525 \text{ Birr}}$$

CHAPTER TWO

REQUIREMENT ANALYSIS

2.1 CURRENT SYSTEM DESCRIPTION

2.1.1 Major function of the current system

The existing internship management system at the University of Gondar operates primarily through a student-led, manual process assisted by paper-based administrative support. The workflow follows a sequential pattern involving the Department, the Student, and the Host Organization:

1. Issuance of Support Letters (Cooper Letter):

The process begins with the University-Industry Linkage (Department Head) office or the Department issuing a generic "Support Letter" (Cooper) to all eligible students. This letter officially requests external organizations to allow the student to conduct their industrial practice.

2. Student-Led Search and Application:

Unlike a centralized matching system, the current burden of finding a placement rests largely on the student. Students physically carry their support letters to various organizations in different cities to inquire about vacancy availability. There is no central database indicating which companies have open slots.

3. Submission of Acceptance Letters:

Once a student finds a willing organization, the company issues a physical "Acceptance Letter." The student must then physically transport this letter back to the Department Coordinator to confirm their placement. The coordinator manually records the student's name and the hosting company in a logbook or Excel sheet.

4. Advisor Assignment and Supervision:

After registration, the department assigns an Academic Advisor to a group of students. The advisor is expected to visit the students at their internship site (Supervision) to monitor progress. However, due to logistical constraints, this monitoring is often limited to a single visit or phone call.

5. Evaluation and Feedback:

At the end of the internship period, the host organization's supervisor fills out a performance

evaluation form. This form is typically placed in a sealed envelope, which the student delivers back to the department. The final grade is calculated based on this industry evaluation (usually 70%) and the university advisor's assessment (usually 30%) after the student presents a report.

2.1.2 Problems of the Existing System

Based on the data collected through interviews with UIL coordinators (Department Head) and direct observation of the current workflow at the University of Gondar, the existing manual system presents several critical challenges that hinder the educational effectiveness of the internship program. These challenges are summarized as follows:

1. Mismatch between Academic Curriculum and Practical Assignment

Unlike professional recruitment, the primary goal of an internship is to reinforce classroom learning through practical experience. However, our assessment of the current manual process revealed that placements are often made based on **alphabetical lists or quota availability** rather than the student's field of study. Consequently, Computer Science students are frequently assigned to organizations where they perform unrelated clerical tasks (such as data entry or paper filing) instead of practicing technical skills like software development or networking. This misalignment defeats the academic purpose of the internship.

2. Limited Access to Internship Opportunities

The reliance on physical notice boards creates a significant information gap. **Student interviews indicated that** high-value internship opportunities are often missed simply because the information is posted only on specific campus bulletin boards. Students who are off-campus or located in different faculty buildings do not receive this information in real-time. Furthermore, the lack of a centralized digital registry makes it difficult for students to verify if a posting comes from a legitimate industry partner recognized by the university.

3. Ineffective Monitoring and Feedback Mechanism

Coordinators admitted that verifying the daily attendance and progress of hundreds of students using paper logbooks is administratively impossible. The current system relies on physical reports submitted at the *end* of the semester, meaning the university has no real-time oversight. If a student stops attending or faces technical challenges during the internship, the department often discovers the issue too late to intervene.

4. Administrative Delays and Manual Dependencies

Our observation of the Department Head office workflow showed that students are required to physically visit the office multiple times to check the status of their recommendation letters. This

manual dependency creates unnecessary queues and delays. Students expressed frustration that significant time is wasted on administrative logistics (getting stamps and signatures) rather than focusing on the internship preparation itself.

5. Lack of Organized Institutional Data

Currently, student internship records and industry feedback forms are stored in disorganized physical folders. **This makes it difficult to retrieve data** for alumni verification or to analyze which companies are providing the best training. The university lacks a digital database to track the long-term effectiveness of the internship program or to maintain a history of industry partnerships.

2.2 REQUIREMENT GATHERING

Requirement Gathering is a pivotal phase where we identify the necessary functionalities and data entities for the proposed system. Instead of focusing on existing problems, this section describes the specific techniques used to extract system requirements and the nature of the data identified during the process.

2.2.1 Requirement Gathering Methods

To accurately define the requirements for the Smart Internship Placement System, the project team utilized a combination of Direct Observation and Document Analysis. Since the development team consists of current students who are direct stakeholders in the system, Participant Observation was also employed to leverage first-hand experience with the existing challenges.

Direct Observation and Participant Observation

Instead of relying on external reports, the project team physically engaged with the daily operations at the University of Gondar Linkage Office to map the current workflow. Through this direct observation, we identified that the placement process follows a fragmented, paper-based trajectory. We observed that students are required to physically visit the office to submit handwritten applications and check notice boards, which helped us identify the need for a "Real-time Status Tracking" feature. Furthermore, as active students within the department, we utilized our own participant experience to validate the difficulty of finding curriculum-aligned internships. This first-hand exposure confirmed that placements are often assigned based on availability rather than technical merit, highlighting the critical need for the "AI-Driven Matching Engine" to align student skills with industry requirements.

2.2.2 Business Rules

This section details the operational policies, constraints, and logical steps that govern the current manual internship coordination process at the University of Gondar. These rules represent the human logic currently applied by coordinators, which will eventually be mapped into the digital logic of the proposed system.

1. Student Eligibility and Registration Rules

- **Academic Standing:** A student is only eligible for an official internship placement if they have a "Clear Status" (no F or NG grades) in all major courses up to the 6th semester.
- **Documentation Requirement:** No student record is considered "Active" until a physical copy of their CV, a copy of their ID, and a recent grade report are submitted and verified by the department coordinator.

- **Application Limit:** In the manual system, a student is restricted to applying for only one official internship placement per semester to prevent double-booking of industry slots.

2. Employer Onboarding and Verification Rules

- **Legal Legitimacy:** Any host organization wishing to admit students for academic internships must provide a valid business license and an official "Letter of Intent" addressed to the University-Industry Linkage (Department Head) Office.
- **Task Alignment:** An internship request is approved only if the technical tasks described by the organization are directly related to the student's field of study (Computer Science) and satisfy the department's curriculum requirements.
- **Stipend Declaration:** Organizations are encouraged to explicitly state any financial support (such as transport or meal allowance) to ensure transparency for the student.

3. The "Managed Brokerage" (Coordinator Logic) Rules

- **Centralized Mediation:** Students are not permitted to bypass the department coordinator. To ensure the internship counts for academic credit, all placements must be formally reported and approved by the coordinator.
- **Profile Forwarding Logic:** The Department Head shall forward student profiles to an employer only after verifying that the student meets the academic prerequisites and that their skills align with the organization's requirements.

4. Placement and Documentation Rules

- **Official Referral:** A placement is only considered "Official" once a hard-copy referral letter is signed by the Linkage Office and stamped with the university seal.
- **Acceptance Confirmation:** The student must return a signed "Acceptance Slip" from the employer within 5 working days of receiving the referral letter; otherwise, the slot is considered vacant and reassigned.

5. Monitoring, Attendance, and Evaluation Rules

- **Logbook Policy:** Students must maintain a daily physical logbook. Each entry must be signed by the industry supervisor. Any logbook with missing signatures or unauthorized alterations is rejected.

- **Supervisor Evaluation:** At the end of the term, the industry supervisor must submit a sealed "Confidential Evaluation Form" directly to the department. If the envelope is opened by the student, the evaluation is voided.

2.3 PROPOSED SYSTEM DESCRIPTION

2.3.1 Overview

The proposed system, titled "Smart Internship Placement for University Students," is an AI-powered, data-driven platform designed to modernize the career linkage process at the University of Gondar. The system transitions the traditional manual internship coordination into a structured "Managed Brokerage" digital platform. It is engineered to eliminate the inefficiencies of random placements by providing an intelligent environment where student competencies are mathematically aligned with industry requirements.

The system primarily functions through three integrated portals: a Student Portal for academic profile curation, an Employer Dashboard for role publication, and a Centralized Admin Panel for university coordinators to maintain institutional oversight. Unlike open job boards, this platform employs a "Human-in-the-Loop" brokerage model, in which university administrators audit and verify every interaction before a placement is finalized.

The system's technical innovation lies in its Intelligent Matching Engine. By utilizing Natural Language Processing (NLP), the platform implements a specialized pipeline:

- **Vector Space Modeling (VSM):** Converting student skills and job descriptions into mathematical vectors using TF-IDF vectorization.
- **Similarity Scoring:** Applying the Cosine Similarity algorithm to calculate a compatibility score, allowing the university to rank and forward candidates **based on their alignment with industry requirements and academic affiliation.**

Furthermore, the system prioritizes data integrity and security. It utilizes JSON Web Tokens (JWT) for secure authentication and Bcrypt hashing for credential protection. By automating administrative tasks such as status notifications, profile mapping, and stipend tracking, the system ensures a transparent, secure, and highly efficient linkage between academia and the industrial sector.

2.3.2 FUNCTIONAL REQUIREMENTS

The functional requirements define the specific behaviors and functions of the system. These requirements describe what the system must do to support the internship placement process for Students, Industry Partners, and the UIL Coordinator (Department Head).

1. User Registration and Authentication

- FR-01: The system shall allow Students to register using their university ID, email, and department details.
- FR-02: The system shall allow Industry Partners to register by providing organizational details, including a business license and contact information.
- FR-03: The system shall allow the UIL Coordinator (Department Head) to verify and approve Industry Partner registrations before they can access the system.
- FR-04: The system shall allow all users to log in securely using their email and password.

2. Student Services

- FR-05 (Profile Management): The system shall allow students to create and update their Persistent Academic Profile, including technical skills, and CV uploads.
- FR-06 (View Opportunities): The system shall display available internship opportunities to students, ranked by relevance based on their profile.
- FR-07 (Apply for Internship): The system shall allow students to submit an internship application for a specific role.
- FR-08 (Track Status): The system shall allow students to view the real-time status of their application (e.g., "Pending Review," "Forwarded," "Accepted," or "Rejected").

3. Industry Partner Services

- FR-09 (Post Vacancy): The system shall allow verified Industry Partners to post internship vacancies with detailed requirements (e.g., Required Skills, Duration, Stipend).
- FR-10 (Manage Vacancies): The system shall allow Industry Partners to edit or close internship postings.
- FR-11 (View Candidates): The system shall allow Industry Partners to view only the student profiles that have been reviewed and forwarded by the UIL Coordinator (Department Head).
- FR-12 (Placement Decision): The system shall allow Industry Partners to accept or reject a forwarded candidate and provide mandatory feedback.

4. Department Head (Brokerage) Services

- FR-13 (Internship Approval): The system shall allow the Department Head to review and approve internship postings to ensure they align with the university curriculum.
- FR-14 (View AI Ranking): The system shall display a list of applicants for each job, ranked by the AI Matching Score (calculated using Cosine Similarity).
- FR-15 (Audit & Forward): The system shall allow the UIL Coordinator (Department Head) to review student applications and manually "Forward" qualified candidates to the Industry Partner.
- FR-16 (Publish Results): The system shall allow the UIL Coordinator (Department Head) to finalize and publish the placement results to students.

5. System Notifications & Reporting

- FR-17: The system shall send automated notifications to students when their application status changes.
- FR-18: The system shall generate statistical reports for the Department Office, showing the number of placements per department and industry partner engagement levels.

2.3.3 NONFUNCTIONAL REQUIREMENTS

Nonfunctional requirements define the operational characteristics and quality attributes of the "Smart Internship Placement System." These requirements ensure that the platform is not only functional but also high-performing, secure, reliable, and user-friendly.

2.3.3.1 Performance

The system shall support real-time AI-based skill matching with minimal latency, ensuring a matching score. This performance is measured and validated using server-side profiling tools (such as Node.js performance hooks) and Chrome Dev Tools to monitor execution time and API response latency. Furthermore, the system shall ensure the instant delivery of automated notifications to students upon application status changes and facilitate quick file uploads for professional photos and CVs over standard network connections. Optimal server response times will be maintained and monitored even during peak recruitment periods using Application Performance Monitoring (APM) tools to ensure the system remains responsive under high traffic.

2.3.3.2 Scalability

The architecture of the system is designed to support the expected workload of the Department of Computer Science (and College of Informatics), accommodating the graduating batch and associated industry partners simultaneously. The system utilizes a modular MERN stack architecture and MongoDB's horizontal scaling capabilities, allowing it to be expanded to other departments in the future without major structural changes."

Scalability is achieved through the horizontal scaling capabilities of the MongoDB database, allowing the system to handle an increasing volume of internship postings and persistent student records without performance degradation. To ensure high traffic management during peak graduation and internship seasons, the system supports load-balancing mechanisms that distribute user requests efficiently across server instances.

2.3.3.3 Availability

The system is designed for high availability, aiming to provide 24/7 accessibility for student internship searches and partner industries internship postings. This requirement ensures that the platform remains functional and accessible to all stakeholders at any time. The availability will be measured and verified during the post-deployment phase by tracking server uptime and recording any instances of system downtime through automated server monitoring tools. To achieve this, the application will be hosted on a reliable cloud infrastructure, and in the event of any system interruption, an automatic recovery process will be triggered to restore services promptly. Scheduled maintenance will be carried out during low-traffic periods to ensure maximum uptime for users.

2.3.3.4 Reliability

The reliability of the platform is ensured through the rigorous logging and timestamping of all recruitment operations, including matching scores, profile forwarding actions, and stipend details, which facilitates auditing. The system shall validate all incoming professional data, such as CV uploads and internship descriptions, to maintain the mathematical integrity of the matching algorithm. To prevent data loss, a redundant cloud backup mechanism is implemented to periodically store student profiles and internship records, ensuring data can be restored in the event of a primary storage failure.

2.3.3.5 Maintainability

Built using modular independent services within the MERN stack, the system ensures high maintainability, allowing the AI matching logic to be updated or refined without disrupting the entire platform. The source code follows standard software development best practices, utilizing tools like ESLint and Prettier, accompanied by detailed technical documentation for future AI enhancements. Advanced error logging tools are integrated to assist administrators in identifying matching logic failures or database connection issues. Furthermore, system configuration settings, such as "Managed Brokerage" rules, are designed to be easily adjustable via the admin panel without requiring direct code changes.

2.3.3.6 Security

The platform enforces strict security protocols, providing each user role (Student, Employer, Admin) with unique login credentials and Role-Based Access Control (RBAC) via JSON Web Tokens (JWT). All sensitive data, including passwords, student contact information, and stipend financial details, is encrypted using Bcrypt and AES-256 standards before storage. Privacy control is strictly enforced, where student contact details remain hidden from employers until the Admin explicitly forwards the profile. Any unauthorized attempts to access the Admin Panel or employer dashboards are logged and reported immediately, and the system enforces automatic session timeouts after prolonged inactivity to ensure data safety.

2.3.3.7 Environmental

The system shall operate as a cross-browser web application, compatible with modern browsers such as Chrome, Firefox, Safari, and Edge, ensuring no specific operating system dependency. The user interface is optimized for both desktop and mobile web browsers, allowing students to check their application status on any device. Additionally, the platform is designed to function effectively even in moderate-bandwidth environments, which is a common infrastructure constraint within university campuses.

2.3.3.8 Usability

The user interface is designed to be intuitive and professional, following modern UI/UX standards such as Material UI or Tailwind CSS, ensuring that industry partners and coordinators can

navigate the system with minimal training. To assist administrators in the selection process, the platform provides clear tooltips and a dedicated help section explaining the AI Matching Score. Detailed documentation and orientation guides will be provided to ensure smooth institutional adoption for all user groups.

2.3.3.9 Interoperability

The platform is designed to be highly compatible with standard data formats, such as JSON for structured data exchange and PDF for professional CV handling, ensuring a seamless flow of information between stakeholders. The system is also capable of integrating with institutional email services to facilitate automated professional communication and status updates. This alignment with standard data exchange protocols ensures that the platform maintains technical compatibility with the university's existing IT infrastructure and potential future institutional databases.

CHAPTER THREE

SYSTEM MODEL

3.1 SCENARIO

A scenario represents a concrete instance of a use case, illustrating a specific sequence of events within the system. In the context of the Smart Internship Placement System, scenarios are used to visualize and understand the system's responses to various user interactions—such as a student discovering an internship through AI suggestions or a UIL Coordinator (Department Head) forwarding a pre-screened profile to an industry partner. These scenarios confirm that the system meets the desired functional requirements and provide a foundation for sequence and collaboration diagrams.

Scenarios are essential because they:

- **Clarify Requirements:** Provide a step-by-step narrative that clarifies how the managed brokerage logic and AI matching should function in a real-world academic setting.
- **Identify Dependencies:** Reveal how the AI similarity score (calculated via TF-IDF) influences the Coordinator's decision to forward a profile.
- **Support Testing:** Serve as the basis for developing test cases to ensure the system accurately ranks students based on technical compatibility.

3.1.1 USE CASE MODEL

A Use Case Model describes how users interact with the Smart Internship Placement System to achieve specific career goals and overcome the challenges associated with manual recruitment and coordination. It provides a high-level blueprint of the system's functional boundaries by outlining:

- **User Objectives:** This defines the specific goals that the primary actors (Students, Industry Partners, and UIL Coordinators (Department Head)) aim to accomplish. These objectives range from students discovering technically aligned internships to employers identifying pre-screened candidates and administrators ensuring institutional oversight.
- **System Interactions:** This details how the platform supports these objectives through its core functionalities. It encompasses features such as persistent academic

profile management, the AI-driven "Smart Finder," and the structured brokerage workflow that allows for controlled communication between students and organizations.

- **System Behavior:** This describes the automated responses and outputs of the system when users perform certain actions. Key behaviors include the real-time generation of similarity scores using TF-IDF and Cosine Similarity algorithms, as well as the triggering of instant status notifications (e.g., Applied, Forwarded, or Accepted) to keep all stakeholders informed.

This model is integral to the development process of the AI-powered marketplace as it forms the foundational framework for design, analysis, documentation, and test case definition. It ensures that every functional requirement—especially the managed brokerage logic—is accurately captured, ensuring that the system delivers a secure, transparent, and intelligent user experience.

Actors of the Smart Internship Placement System

The operating environment of the system is defined by a synergistic interaction between human actors, the Natural Language Processing (NLP) matching engine, and the structured recruitment data. To maintain institutional oversight and ensure high-precision matching, the system identifies three primary actors:

UIL Coordinator (Department Head)

The University-Industry Linkage (Department Head) Coordinator acts as the central intermediary who governs the placement lifecycle. This actor ensures that the transition from academia to industry is both verified and curriculum-aligned.

Key Roles and Responsibilities:

- **Industry Partner Verification:** Validates registration requests from external organizations to ensure they are legitimate partners capable of providing quality industrial exposure.
- **Managed Brokerage Control:** Serves as a gateway for student applications, reviewing all submissions before they are permitted to reach the employer.
- **AI Matching Oversight:** Interfaces with the AI Engine to monitor similarity scores, ensuring that student technical competencies accurately align with the specific internship requirements.
- **Profile Forwarding:** Securely transfers pre-screened student academic identities to industry partners for the final selection phase.
- **Decision Publication:** Finalizes the recruitment process by publishing official results, making the outcome visible to both students and host organizations.
- **Strategic Reporting:** Analyzes system-wide analytics, including placement trends and technical skill gaps, to provide data-driven insights for the university.

Expectation: The UIL Coordinator (Department Head) expects a robust, centralized command center that streamlines candidate screening via AI ranking and provides granular control over the managed brokerage workflow.

Student

The Student is the primary beneficiary of the system, seeking an industrial placement that matches their specific academic background and technical skillset.

Key Roles and Responsibilities:

- **Persistent Academic Identity Management:** Constructs and maintains a verified professional profile, including academic records (GPA), standardized technical tags, and project portfolios that persist across multiple applications.
- **AI-Driven Opportunity Discovery:** Utilizes the "Smart Finder" feature to browse internship roles that are automatically ranked based on mathematical similarity to their profile.
- **Broker-Mediated Application:** Submits targeted applications for specific roles, which are initially routed to the University Admin for quality auditing.
- **Real-time Status Tracking:** Monitors the progression of their application through an automated timeline (e.g., Applied → Under Review → Forwarded → Accepted).

Expectation: The Student expects a transparent, intelligent, and user-centric platform that reduces placement uncertainty by providing relevant internship suggestions and real-time feedback.

Industry Partner (Employer)

The Industry Partner represents the external organization providing the placement opportunity and seeking technically competent students.

Key Roles and Responsibilities:

- **Technical Requirement Specification:** Configures detailed internship listings, defining specific tools, frameworks, and stipend details required for the role.
- **Vetted Candidate Review:** Accesses a curated pool of high-quality student profiles that have been exclusively pre-screened and forwarded by the university administration.
- **Technical Profile Access:** Evaluates the persistent profiles and academic histories of forwarded students to make informed hiring decisions.
- **Recruitment Decision-making:** Submits formal "Approve" or "Reject" responses back to the Admin, ensuring a closed-loop communication flow.

Expectation: The Employer expects a high-efficiency interface that provides access to a technically aligned talent pool, significantly reducing the administrative burden of manual candidate filtering.

Use Case Identification

The functional boundaries of the Smart Internship Placement System are defined through a set of core use cases that facilitate an intelligent, broker-managed placement process. These use cases are categorized based on their scope and the specific strategic objectives of the stakeholders involved.

System-Wide Use Cases (Cross-Functional)

These functionalities provide the underlying security and management framework accessible to all authenticated actors.

1. **Secure User Authentication (JWT-Based):** Validates user identity through unique credentials and issues JSON Web Tokens for stateless session management.
2. **Global Session Termination:** Destroys active session tokens to prevent unauthorized access upon completion of system tasks.

3. **Institutional/Personal Profile Management:** Enables users to curate and update their primary identity data, ensuring the persistence of contact information and credentials.

UIL Coordinator (Department Head) Use Cases

The Coordinator acts as the central quality controller, ensuring the integrity of the university-industry linkage.

1. **Industry Partner Accreditation (Verification):** A rigorous process of auditing and approving registration requests from host organizations to ensure institutional legitimacy.
2. **NLP-Driven Match Analysis:** Reviewing the similarity scores generated by the AI Engine (TF-IDF/Cosine Similarity) to identify the most technically compatible candidates.
3. **Managed Application Brokerage (Forwarding):** The exclusive action of transferring high-ranking student profiles to industry partners, acting as a human-in-the-loop filter.
4. **Formal Result Publication:** Finalizing the placement lifecycle and transitioning the application status to "Accepted," which triggers official documentation.
5. **Strategic Linkage Analytics:** Generating comprehensive reports on placement statistics, stipend trends, and industrial engagement levels for academic auditing.
6. **Granular User Governance:** Managing and monitoring user accounts to maintain system safety and compliance.

Industry Partner (Employer) Use Cases

These use cases allow external organizations to interact with the university's talent pool in a controlled environment.

1. **Internship Requirement Modeling (Posting):** Defining detailed technical specifications, durations, and stipend structures for available placement slots.
2. **Vetted Candidate Evaluation:** Accessing and reviewing a curated list of student profiles that have been pre-screened and forwarded by the Department office.
3. **Recruitment Decision Feedback:** Submitting formal "Approve" or "Reject" responses back to the brokerage dashboard for final administrative review.

4. **Vacancy Lifecycle Management:** Monitoring and adjusting active job listings, including editing requirements or closing fulfilled roles.

Student Use Cases

The Student uses the system as a discovery and professional identity platform.

1. **Persistent Academic Identity Construction:** Building a robust digital profile that includes verified technical skills, academic performance (GPA), and project history.
2. **AI-Enhanced Opportunity Discovery (Smart Finder):** Utilizing an NLP-powered search engine that prioritizes internship suggestions based on their alignment with the student's profile.
3. **Broker-Mediated Placement Application:** Submitting targeted applications that are systematically routed through the Coordinator(Department Head) for initial screening.
4. **Real-time status monitoring:** Tracking the progress of applications through a transparent notification system (e.g., Applied → Under Review → Forwarded → Accepted).

Use Case Scenario

Use Case Scenario 1: UIL Coordinator (Department Head)

Scenario Name: - Institutional Oversight and Application Brokerage Management

Primary Actor: - UIL Coordinator (Department Head)

Pre-condition: The Coordinator must have an active administrative account with authorized Role-Based Access Control (RBAC) permissions.

Flow of Events:

1. Secure Authentication and Command Center Access:

- ❖ The UIL Coordinator (Department Head) navigates to the centralized management gateway.
- ❖ The Coordinator inputs administrative credentials (Email and encrypted Password).
- ❖ The system validates the identity via **JWT (JSON Web Tokens)** and checks for specific administrative privileges in the database.
- ❖ Upon successful validation, the system redirects the actor to the **Admin (Department Head) Command Center**, a dashboard designed for real-time institutional oversight.

2. Industry Partner Accreditation and Verification:

- ❖ The Department Head selects the **"Partner Management"** module to review pending registration requests from external organizations.
- ❖ The Department Head audits institutional details (e.g., business license, mission alignment) to ensure the industry partner can provide a quality educational environment.
- ❖ Upon verification, the Department Head approves the account, granting the organization the privilege to post curriculum-aligned internship opportunities.

3. NLP-Driven Application Audit and Match Review:

- ❖ The Department Head navigates to the **"Brokerage Dashboard"** to review active internship postings and their respective applicants.
- ❖ For a specific role, the system presents a prioritized list of students, ranked by the **AI Matching Engine**.
- ❖ The Department Head analyzes the **Match Score (%)**, which is dynamically generated using **TF-IDF vectorization and Cosine Similarity** to compare student skills against role requirements.

- ❖ The Department Head cross-references the AI score with the student's persistent academic profile (verified GPA, technical projects, and course history).

4. Managed Brokerage Execution (Profile Forwarding):

- ❖ Based on the AI-ranked data and academic standing, the Department Head identifies the most technically compatible candidates.
- ❖ The Department Head triggers the **"Forward to Industry Partner"** action.
- ❖ The system securely transfers the student's academic identity to the specific industry partners dashboard, effectively acting as a human-in-the-loop quality filter.
- ❖ The system transitions the application status to **"Forwarded"** and triggers an automated, real-time notification to the student.

5. Stipend Transparency and Linkage Monitoring:

- ❖ The Department Head utilizes the **Financial Transparency Module** to track stipend declarations and ensure they meet university standards.
- ❖ The system monitors the communication logs between the student and the industry partner to ensure a smooth transition.
- ❖ The Department Head reviews key performance indicators (KPIs) such as:
 - Cumulative number of active industry partnerships.
 - Placement success rates per department.
 - Average accuracy and alignment of the AI matching logic.

6. Final Outcome Publication and Session Termination:

- ❖ Once the industry partner submits a hiring decision, the Department Head reviews the feedback.
- ❖ The Department Head performs the **"Publish Result"** action, finalizing the placement record in the database.
- ❖ The system sends an official "Placement Confirmed" notification and generates the formal digital recommendation letter for the student.

- ❖ To ensure security, the Department Head clicks the **Logout** button; the system destroys the JWT session token and redirects to the login landing page.

Use Case Scenario 2: Industry Partner (Employer)

Scenario Name: - Internship Opportunity Modeling and Vetted Candidate Selection

Primary Actor: - Industry Partner (Employer)

Pre-condition: - The Industry Partner must have an Admin-verified account and an active session established via the secure authentication gateway.

Flow of Events:

1. Authenticated Access and Partner Analytics:

- ❖ The Industry Partner logs into the system using their accredited credentials.
- ❖ The system validates the session via **JWT** and redirects the user to the **Employer Dashboard**.
- ❖ The Partner reviews real-time recruitment statistics, including the status of active internship postings and the number of candidates currently being pre-screened by the University Admin.

2. Industrial Requirement Modeling (Opportunity Creation):

- ❖ The Partner selects the "**Create Internship Opportunity**" module.
- ❖ The system provides a structured form where the Partner inputs critical data points, including:
 - **Technical Specifications:** Detailed descriptions of required tools (e.g., React, SQL), frameworks, and project scopes. This text is utilized by the **NLP Engine** for similarity vectorization.
 - **Logistical Details:** Internship duration (e.g., 3-6 months), department category, and physical/remote location.

- **Stipend Transparency:** Explicit declaration of financial compensation or student allowances to ensure institutional compliance.

The system validates the input for completeness and submits the posting to the **UIL Coordinator (Department Head)** for academic auditing and approval.

3. Vetted Candidate Evaluation (Reviewing Forwarded Profiles):

- ❖ Upon receipt of an automated notification that the Admin (Department Head) has "Forwarded" qualified candidates, the Partner accesses the "**Curated Talent Pool**" section.
- ❖ The Partner evaluates a list of student profiles that have been exclusively pre-screened for both technical and academic alignment.
- ❖ The Partner analyzes the **AI Matching Score (%)** for each candidate, gaining insight into how well the student's skills align with the specific project requirements.
- ❖ The Partner reviews the **Persistent Academic Profile**, which includes verified technical project links, portfolios, and CGPA.

4. Recruitment Decision-making and Feedback Loop:

- ❖ The Partner performs a comparative analysis of the forwarded candidates based on the AI ranking and academic history.
- ❖ The Partner selects the "**Approve**" or "**Reject**" action for each candidate.
- ❖ For each decision, the system requires mandatory feedback, which is routed back to the UIL Coordinator (Department Head) to maintain a transparent brokerage workflow.
- ❖ If a candidate is approved, the Partner gains access to the student's contact information (once the Admin (Department Head) publishes the final result) to finalize technical interviews or onboarding logistics.

5. Vacancy Lifecycle Management and Reporting:

- ❖ The Partner monitors the status of their postings, with the ability to **edit** requirements or **Close** vacancies once the placement goals are achieved.
- ❖ The system generates summary reports for the Partner, detailing the efficiency of the placement cycle and the technical quality of the applicants.

6. Secure Session Termination:

- ❖ Once the recruitment tasks are finalized, the Partner initiates the **Logout** sequence.
- ❖ The system neutralizes the active JWT session token and redirects the actor to the system landing page, ensuring the security of institutional data.

Use Case Scenario 3: Student

Scenario Name: - Persistent Academic Identity Management and AI-Driven Internship Discovery

Primary Actor: - Student

Pre-condition: - The Student must be a registered member of the university with valid credentials to access the placement portal

Flow of Events:

1. Secure Authentication and Dashboard Initialization:

- ❖ The Student navigates to the system's centralized login gateway.
- ❖ The Student inputs their unique credentials (Email and Password).
- ❖ The system validates the identity via **JWT (JSON Web Tokens)** and retrieves the user's specific role permissions.
- ❖ Upon successful authentication, the system redirects the Student to the **Student Dashboard**, which presents a real-time overview of placement statistics, active system notifications, and curriculum-aligned internship suggestions.

2. Persistent Academic Identity Construction:

- ❖ The Student accesses the **"My Academic Profile"** module.

- ❖ The Student populates their digital identity with critical data points:
 - **Technical Skillsets:** Standardized tags for programming languages, frameworks, and tools (e.g., Python, React, TensorFlow).
 - **Project Portfolio:** Links to technical projects (e.g., GitHub) and documented academic achievements.
 - **Official Records:** Uploads of digital transcripts (GPA) and a formal identification photo.
- ❖ The system persists this data in the **MongoDB database**, ensuring it is automatically attached as a "verified resume" to all future internship applications, eliminating redundant data entry.

3. Intelligent Opportunity Discovery (The Smart Finder):

- ❖ The Student navigates to the "**AI Smart Finder**" section.
- ❖ The system initializes the placement environment, which includes:
 - **Contextual Search:** Ability to filter by department, duration, and stipend availability.
 - **AI-Ranked Suggestions:** The engine executes a **TF-IDF and Cosine Similarity** analysis to compare the Student's skills against active internship requirements.
- The Student reviews a list of internship roles ranked by a **Match Score (%)**, providing immediate insight into their technical compatibility with host organizations.

4. Managed Placement Application:

- ❖ The Student selects a specific internship role and triggers the "**Apply**" action.
- ❖ The system verifies that the **Persistent Academic Profile** is complete.
- ❖ **Brokerage Logic Execution:** The application is systematically routed to the **UIL Coordinator's dashboard** for initial institutional screening. The application remains invisible to the Industry Partner at this stage to maintain managed oversight.

- ❖ The system logs the entry as "**Applied**" and updates the Student's real-time tracking timeline.

5. Placement Lifecycle Monitoring and Engagement:

- ❖ The Student utilizes the **Notification Center** to monitor the progress of their application through various states:
 - **Under Review:** The Admin (Department Head) is auditing the academic alignment.
 - **Forwarded:** The profile has been vetted and sent to the Industry Partner.
 - **Final Decision:** The result is published (Accepted/Rejected).
- ❖ The Student receives automated alerts if an Industry Partner requests a technical interview or provides feedback via the UIL Coordinator (Department Head).

6. Secure Session Termination:

- ❖ Upon completion of system interactions, the Student initiates the **Logout** sequence.
- ❖ The system prompts for confirmation to prevent data loss, then destroys the active JWT session token.
- ❖ The Student is redirected to the login landing page, ensuring the privacy of their academic records.

3.1.2 USE CASE DIAGRAM

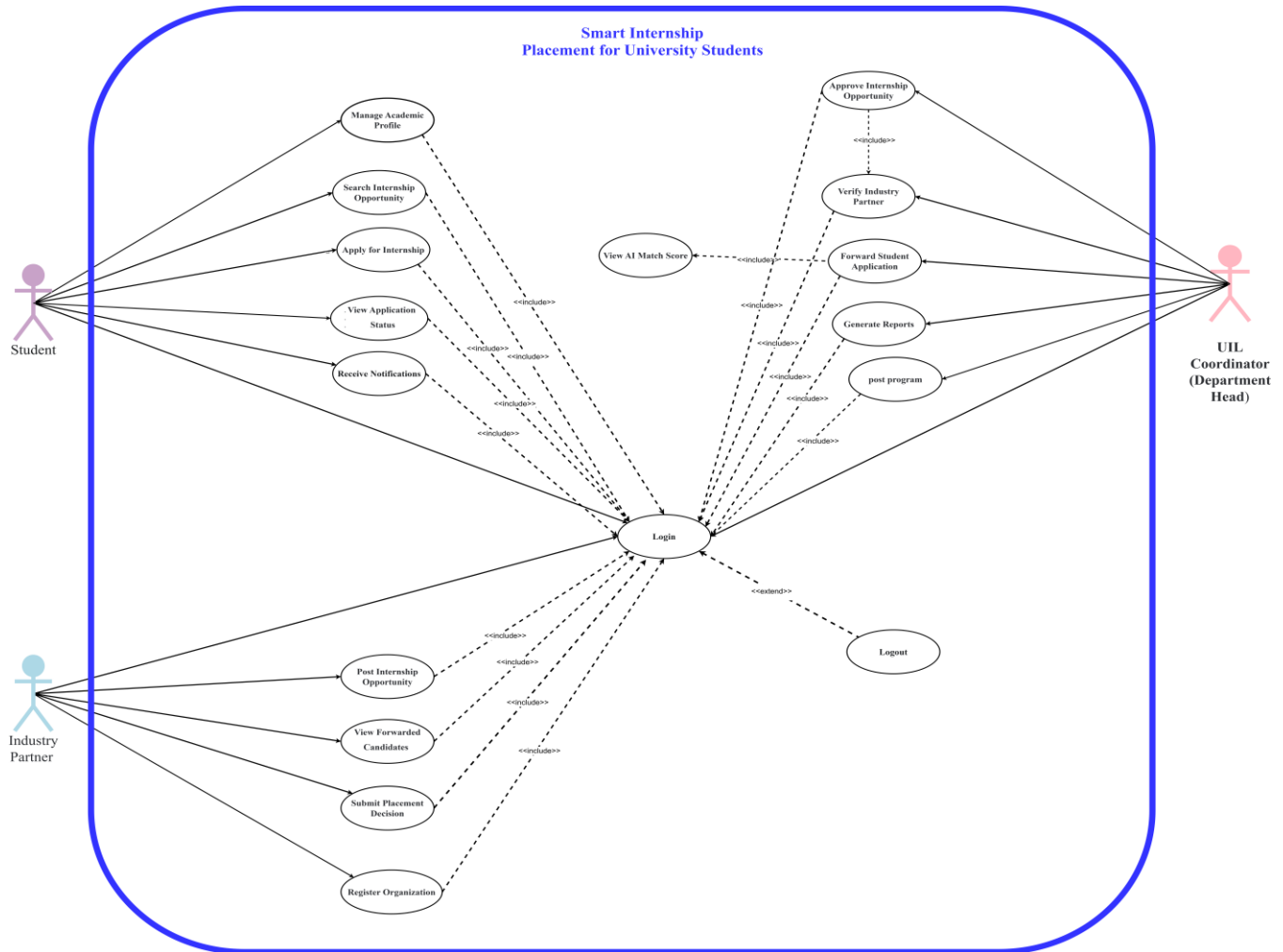


Figure 3-11 Use Case Diagram

3.1.3 DESCRIPTION OF USE CASE MODEL

Use Case Description

The Use Case Description section provides a detailed explanation of how different users interact with the Smart Internship Placement for University Students. Each use case outlines a specific functionality of the system, describing the interactions between key actors (Students, Industry Partners, and UIL Coordinators(Department Head)) and the system to achieve a particular objective, such as creating a persistent profile, posting an opportunity, or executing the managed brokerage workflow.

This section includes key elements such as:

- **Use Case Name** – The title of the use case that defines the functionality.
- **Description** – A brief explanation of the purpose and scope of the use case.
- **Actors** – The users or external systems that participate in the interaction.
- **Preconditions** – The conditions that must be met before the use case starts.
- **Postconditions** – The state of the system after the use case execution.
- **Basic Course of Action** – The standard flow of events in the use case.
- **Alternative Course of Action** – Deviations or exception scenarios that may occur (e.g., rejection of a candidate or invalid data entry).

These descriptions offer a structured overview of the system's functionalities, ensuring that all interactions—particularly the AI matching logic and institutional oversight protocols are well-documented and understood. This is essential for system design, development, and validation, as it helps in identifying user needs, system responses, and potential improvements.

Use Case Name	User Login
Description	This Use Case allows all system users (Student, Industry Partner, UIL Coordinator (Department Head)) to authenticate into the Smart Internship Placement System by providing valid credentials. The system redirects them to their role-specific dashboard.
Actors	Student, Industry Partner (Employer), UIL Coordinator (Department Head)
Precondition	1. The user must have a registered account in the system.

	2. The system database must be accessible.
Postcondition	1. The user is authenticated and granted access to their specific portal. 2. A secure session (JWT) is established. 3. The system logs the login time for security auditing.
Basic Course of Action	1. The user navigates to the central login page. 2. The user enters their registered Email and Password. 3. The system validates the credentials against the MongoDB database. 4. If credentials are valid, the system retrieves the user role. 5. The system redirects the user based on their role: ▪ Student → Student Dashboard (Smart Finder). ▪ Industry Partner → Employer Portal. ▪ Department Head → Central Brokerage/Admin Panel. 6. The system logs the successful login activity.
Alternative Course of Action	1. Invalid Credentials: If the email or password is incorrect, the system displays an error message and prompts the user to retry. 2. Unverified Account (Industry Partner):

	If an Industry Partner attempts to log in but hasn't been approved by the UIL Coordinator yet, the system denies access and displays "Account Pending Verification". 3. Account Locked/Suspended: If the UIL Coordinator has suspended an account for policy violations, access is denied.
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Table 3-1: Use Case Description for User Login

Use Case Name	Post Internship Opportunity
Description	This use case allows an authenticated Industry Partner (Employer) to create a new internship vacancy. The posting includes detailed job descriptions and required technical skills (used for AI matching), as well as stipend information. The post is submitted to the UIL Coordinator for approval.
Actors	Industry Partner (Employer), UIL Coordinator (Admin)
Precondition	1. The Industry Partner must be logged into the system. 2. The Industry Partner's account must be verified by the UIL Coordinator.
Postcondition	1. A new opportunity record is created with the status "Pending Approval".

	2. The UIL Coordinator receives a notification to review and approve the posting. 3. The AI Engine prepares to index the job description for matching.
Basic Course of Action	1. The Industry Partner navigates to the "Create Opportunity" section on their dashboard. 2. The system displays a form requiring: Job Title, Detailed Description, Required Technical Skills (for NLP analysis), Duration, and Stipend Amount. 3. The Partner fills in the details and clicks "Submit for Approval". 4. The system validates that all mandatory fields are populated. 5. The system saves the posting in the database with the status "Pending Approval". 6. The system sends an automated notification to the UIL Coordinator for review.
Alternative Course of Action	1. Invalid Input / Missing Data: If mandatory fields (especially "Required Skills" needed for AI or "Stipend" needed for transparency) are empty, the system displays an error message. 2. Unverified Account: If the Industry Partner's account status is still "Pending Verification," the system

	denies access to the posting feature and prompts them to contact the UIL office.
--	--

Table 3-2: Use Case Description for Post Opportunity

Use Case Name	Apply for an Internship
Description	This use case allows a Student to apply for an open internship position found via the Smart Finder. The system automatically retrieves their Persistent Academic Profile, calculates the AI Matching Score using NLP, and routes the application to the UIL Coordinator for initial screening.
Actors	Student, System (AI Engine)
Precondition	1. The Student must be logged into the system. 2. The Student must have a completed Persistent Academic Profile (CV, Skills, GPA).
Postcondition	1. The application is submitted to the UIL Coordinator's dashboard with the status "Under Review". 2. The AI Match Score is recorded and

	attached to the application. 3. The Student receives a confirmation notification.
Basic Course of Action	1. The Student navigates to the "Smart Finder" to view AI-ranked opportunities. 2. The Student selects a specific internship and clicks the "Apply" button. 3. The System retrieves the student's persistent profile and executes the NLP Algorithm to calculate the similarity score against the job requirements. 4. The System attaches the profile, CV, and the calculated Match Score to the application. 5. The System routes the application to the UIL Coordinator (it remains invisible to the Employer at this stage). 6. A confirmation message, "Application Sent for Review," is displayed to the Student.
Alternative Course of Action	1. Incomplete Profile: If the student's profile (e.g., CV or Skills) is missing, the system redirects them to the "Edit Profile" page and prevents the application.

	2. Deadline Passed: If the internship posting deadline has expired, the "Apply" button is disabled. 3. Duplicate Application: If the student has already applied for this role, the system displays "You have already applied for this position."
--	---

Table 3-3: Use Case Description for Apply for Internship

Use Case Name	Forward Student Profile (Brokerage)
Description	This use case describes the core brokerage logic where the UIL Coordinator reviews student applications, checks the AI Matching Score, and forwards the best-fit candidates to the Industry Partner.
Actors	UIL Coordinator (Admin), Industry Partner (Employer)

Precondition	1. Students must have applied for the internship. 2. The UIL Coordinator must be logged in.
Postcondition	1. The selected student profile becomes visible to the Employer. 2. The application status updates to "Forwarded." 3. The Student receives a notification.

Basic Course of Action	1. The Admin navigates to the "Application Review" dashboard. 2. The system displays applicants ranked by their AI Matching Score. 3. The Admin reviews the top candidates' profiles. 4. The Admin selects qualified students and clicks "Forward to Industry Partner". 5. The system grants the Employer access to view these specific profiles. 6. The system sends a notification to the Employer and the Student.
Alternative Course of Action	1. Rejection: If the student does not meet the requirements, the Admin clicks "Reject". The status updates to "Rejected," and the profile is <i>never</i> shown to the Employer.

Table 3-4: Use Case Description for the Brokerage Logic

3.1.4 ACTIVITY DIAGRAM

The Activity Diagram is a key behavioral UML diagram that illustrates the system's dynamic workflow. It visually represents how activities flow from one step to another, showing the interactions between Students, Employers(industry partner), UIL Coordinators (Department Head), and the System. These diagrams use Swimlane to differentiate the responsibilities of each actor, grouping activities based on their involvement.

Each Use Case previously identified corresponds to a specific workflow, which will be detailed using Activity Diagrams. These diagrams help in understanding the sequential flow of activities, decision points, and interactions within the system.

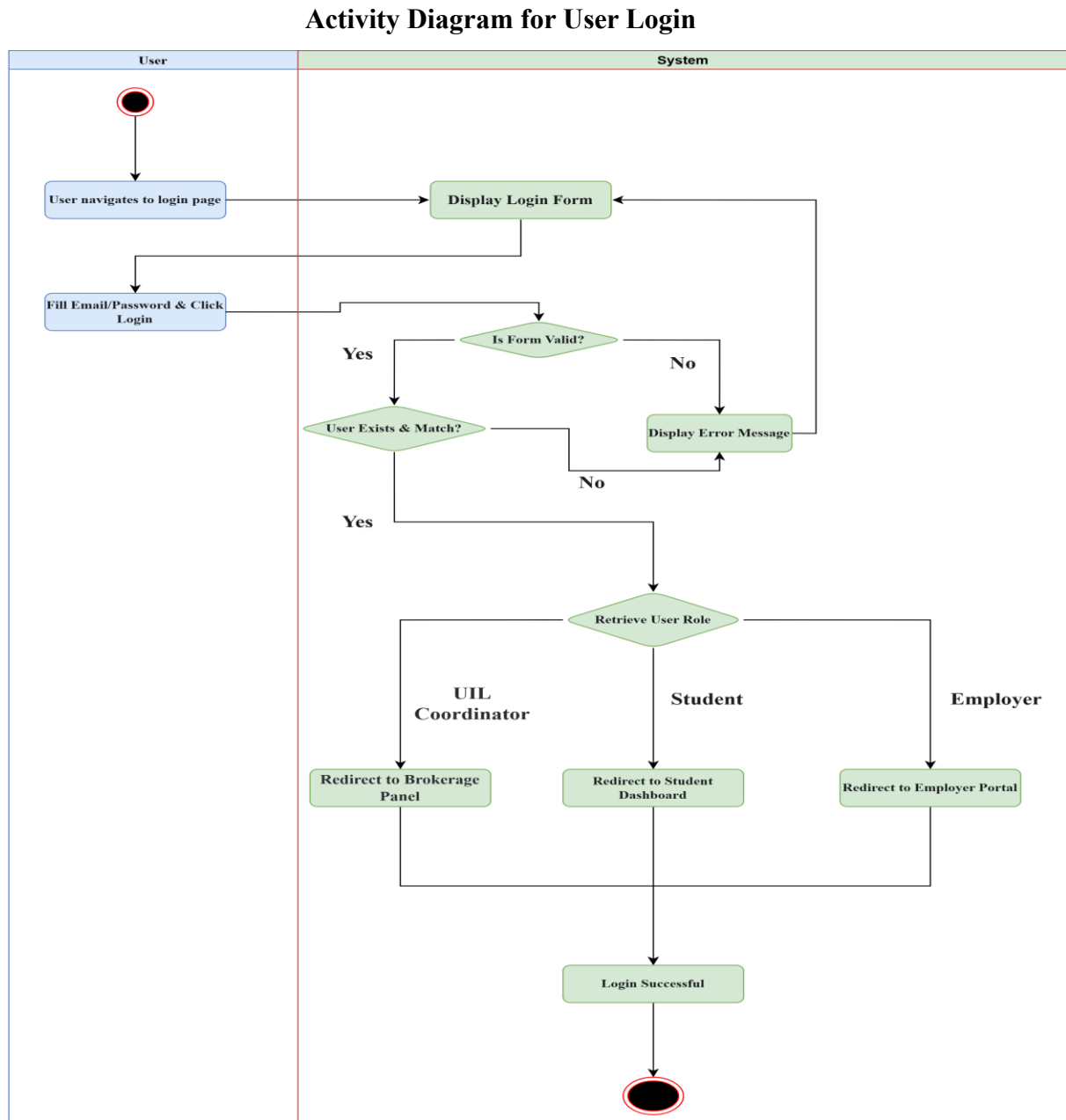


Figure 3-12 User Login Activity Diagram

Description of User Login

1. Login Initiation:

User: Navigates to the login page from the system landing page.

System Response: Displays the Login Form to the user.

2. Enter Credentials & Validation Check:

User: Fills in the Email and Password fields and clicks the "Login" button.

System Response (Is Form Valid?): The system first checks if the input format is correct and fields are not empty.

- *If No:* The system displays an error message and redirects back to the Login Form.
- *If Yes:* The system proceeds to verify the credentials.

3. Database Verification (User Exists & Match?):

System Response: The backend validates the credentials against the database.

- *If No (Invalid Credentials):* The system displays an error message prompting the user to retry, and redirects back to the "Display Login Form" step.
- *If Yes:* The system proceeds to retrieve the user's role.

4. Role-Based Redirection (Retrieve User Role):

System Response: The system identifies the user's specific role and redirects them accordingly:

- Student: Redirects to the Student Dashboard.
- Employer (industry partner): Redirects to the Employer Portal.
- UIL Coordinator (Department Head): Redirects to the Brokerage Panel.

5. Login Success:

System Response: The user is successfully authenticated, and the specific dashboard/portal is loaded.

User: Access granted and session starts.

ACTIVITY DIAGRAM FOR PROGRAM SETUP AND OPPORTUNITY POSTING

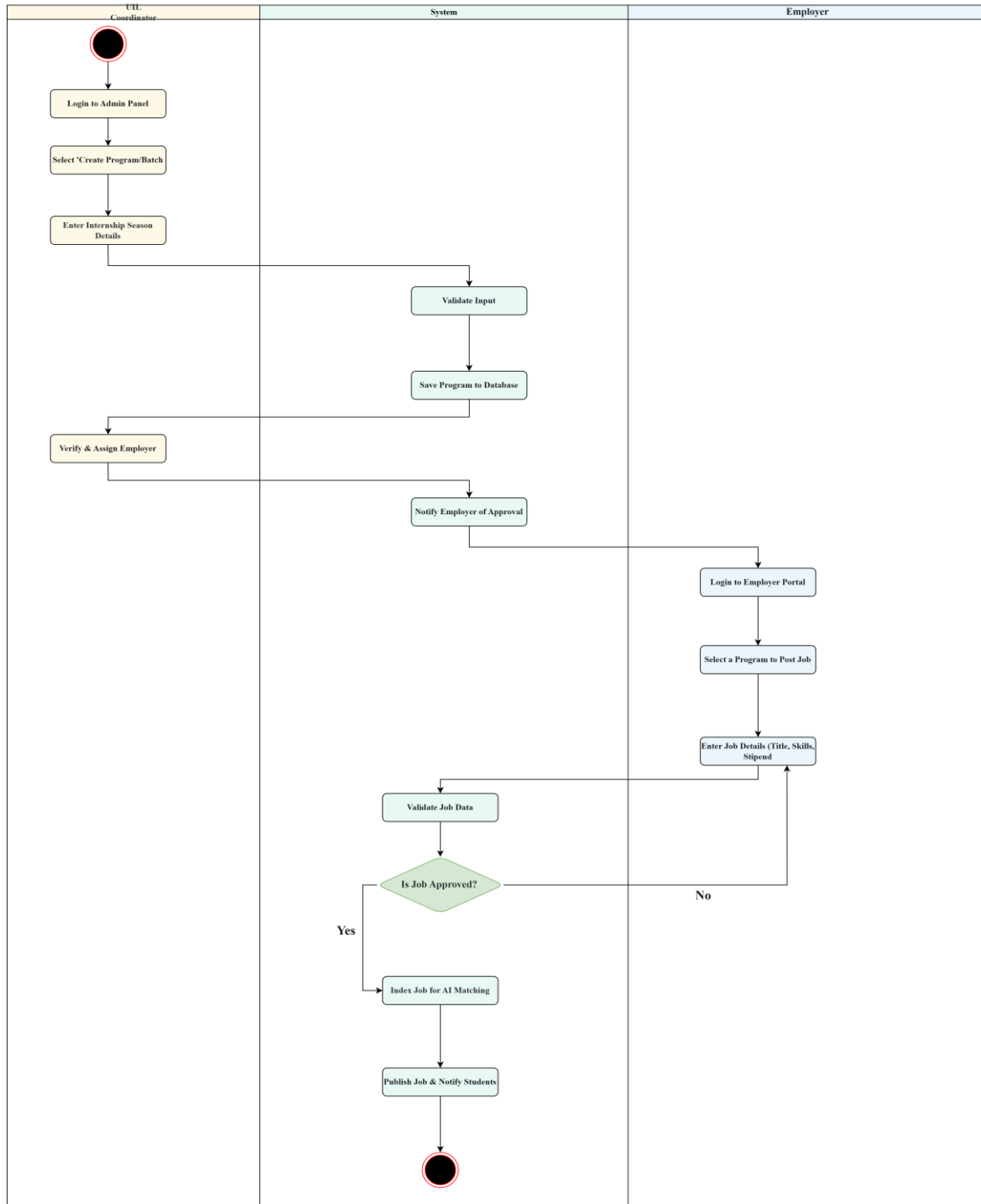


Figure 3-13: Activity Diagram for Program Setup and Opportunity Posting

Description of Activity Diagram for Program Setup and Opportunity Posting

This activity diagram illustrates the sequential workflow required to initialize an internship season and publish internship opportunities. It highlights the interaction between the UIL Coordinator (Department Head), the System Backend, and the Industry Partner (Employer). The process ensures that internship roles are only posted within a valid, university-approved program cycle.

1. Program Initialization (UIL Coordinator (Department Head) Action)

1.1 Log in to Admin Panel:

- **Actor:** UIL Coordinator (Admin)
- **Action:** The Coordinator logs into the secure admin panel to access the "Program Management" module.
- **System Response:** Authenticates the user and loads the dashboard.

1.2 Select 'Create Program/Batch':

- **Actor:** UIL Coordinator (Department Head)
- **Action:** Initiates the creation of a new internship cycle (e.g., "Summer 2026 Batch" or "Graduating Class Internship").

1.3 Enter Internship Season Details:

- **Actor:** UIL Coordinator (Department Head)
- **Action:** Inputs critical constraints for the season, including:
 - ❖ *Start/End Dates:* The valid duration for the internship.
 - ❖ *Eligible Departments:* Which students (e.g., CS, IT) can apply.
 - ❖ *Minimum Duration:* e.g., 45 days.

1.4 Verify & Assign Employer:

- **Actor:** UIL Coordinator (Department Head)
- **Action:** The Coordinator selects active, verified Industry Partners and assigns them permission to post opportunities under this specific program batch. This enforces the "Managed Brokerage" control.

2. System Validation and Notification (System Logic)

2.1 Validate Input:

- **Action:** The system checks if the program dates and details are logical and complete.

2.2 Save Program to Database:

- **Action:** If valid, the new Program Batch is committed to the MongoDB database.

2.3 Notify Employer of Approval:

- **Action:** The system automatically sends a notification to the assigned Industry Partners, informing them that the portal is now open for posting requirements.

3. Opportunity Posting (Industry Partner Action)

3.1 Login to Employer Portal:

- **Actor:** Industry Partner (Employer)
- **Action:** The partner logs in using their verified credentials.

3.2 Select a Program to Post Job:

- **Actor:** Industry Partner
- **Action:** The partner selects the active program (e.g., "Summer 2026") enabled by the UIL Coordinator (Department Head).

3.3 Enter Job (Internship) Details:

- **Actor:** Industry Partner
- **Action:** The partner fills out the "Opportunity Creation Form". Key data points include:
 - ❖ *Title:* e.g., "Frontend Developer Intern".
 - ❖ *Required Skills:* Technical keywords (React, Node.js) crucial for the **AI Matching Engine**.
 - ❖ *Stipend:* Financial details for transparency.

4. Opportunity Validation and AI Indexing (System Logic)

4.1 Validate Job Data:

- **Action:** The system ensures all mandatory fields (especially Skills and Stipend) are filled.

4.2 Is Job Approved? (Decision Node):

- **Action:** The system checks against the business rules (e.g., Is the account verified? Is the stipend field populated?).

- *If No:* The posting is rejected, and the Employer is prompted to correct errors.
- *If Yes:* The process proceeds to indexing.

4.3 Index Job for AI Matching:

- **Action:** This is a critical technical step. The **NLP Engine** processes the job description and skills, converting them into vectors (TF-IDF) to prepare for similarity calculations with student profiles.

4.4 Publish Internship & Notify Students:

- **Action:** The opportunity becomes visible on the marketplace. The system sends real-time notifications to eligible students whose academic profiles match the indexed skill requirements.

ACTIVITY DIAGRAM FOR AI-DRIVEN SMART MATCHING AND APPLICATION WORKFLOW

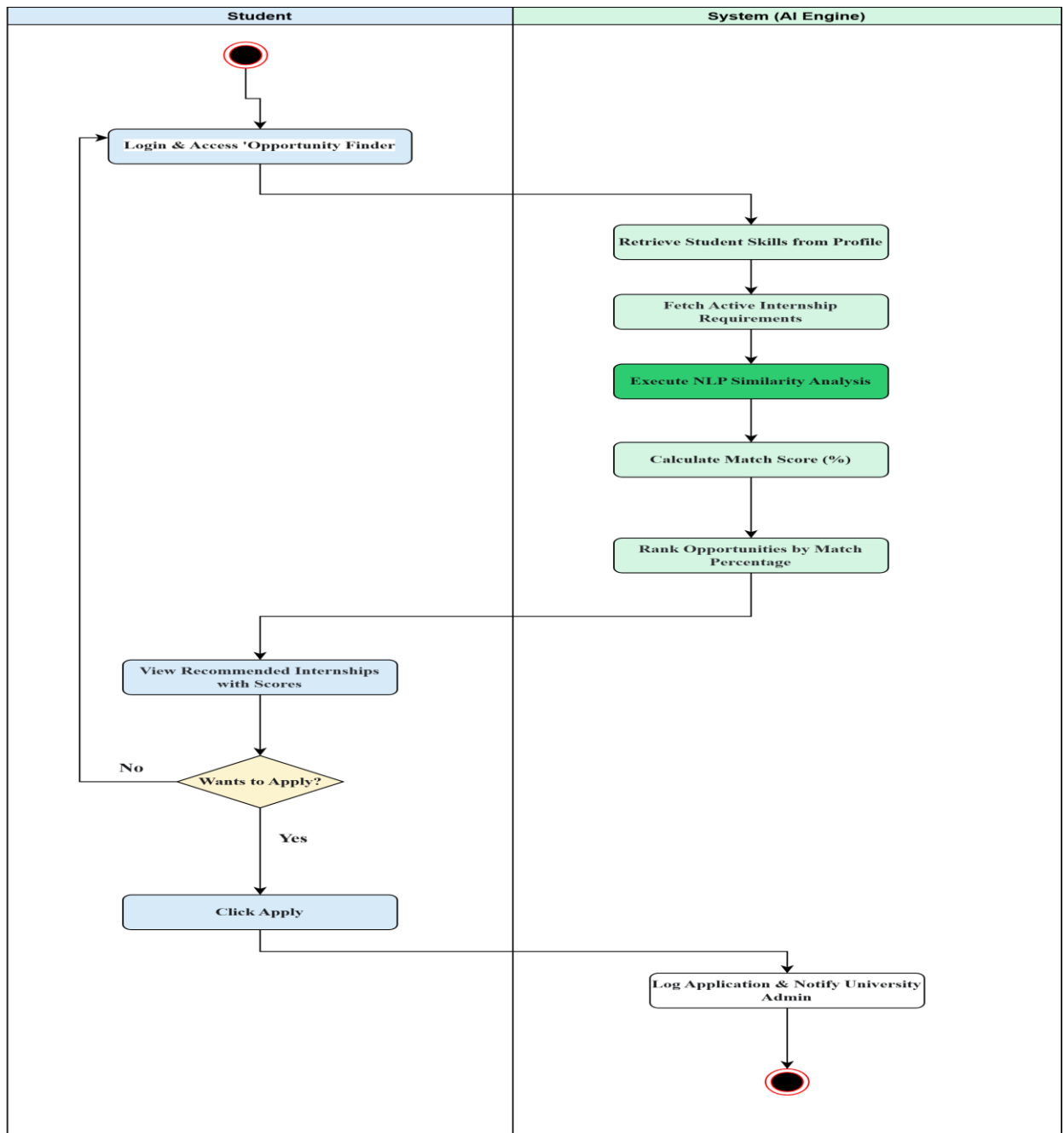


Figure 3-14: Activity Diagram for AI-Driven Smart Matching and Application Workflow

Description for AI Smart Matching Activity Diagram

1. Data Retrieval Phase:

- **Student:** After logging in, the student accesses the 'Opportunity Finder' section.
- **System (AI Engine):** Automatically scans and retrieves technical skills (e.g., Programming languages, frameworks) from the student's Persistent Academic Profile stored in the MongoDB database.
- **System:** Simultaneously, it fetches all active internship requirements posted by verified Industry Partners.

2. Intelligent Processing (NLP):

- **AI Analysis:** The system executes a Natural Language Processing (NLP) similarity algorithm. It compares the technical keywords of the student's skills against the internship descriptions to find the best academic alignment.
- **Scoring:** Based on the analysis, the engine calculates a Match Score (%) for each opportunity.
- **Ranking:** The system ranks all matching opportunities in descending order, displaying the most relevant internships at the top of the student's list.

3. Student Interaction and Application:

- **Student:** Reviews the recommended internships and their respective match scores.
- **Decision (Wants to Apply?):**
 - **No:** The student continues browsing the finder.
 - **Yes:** The student clicks the "Apply" button for the specific role.

4. Final Submission:

- **System:** Formally logs the application, attaches the AI Match Score, and sends a real-time notification to the University Admin (Broker) for final screening and forwarding to the Industry Partner.

3.1.5 OBJECT MODEL

The Object Model above illustrates a runtime snapshot of the AI-powered managed system. It depicts how specific data instances interact within the system memory during an active placement session.

In this instance, student1 (Aklile Tsege) has submitted an application for internship1 (Full Stack Developer), which falls under the program1 category. The industry partner, Partner 1 (Tech Solutions Ltd), is the verified host organization for this specific internship opportunity.

A key element in this model is application1, which acts as the associative entity connecting the student to the internship. It stores critical linkage data, including the AI-generated match score and the application status ("Forwarded"). This status indicates that the University Admin has reviewed the match and officially forwarded the student's profile to the industry partner, strictly adhering to the managed brokerage workflow.

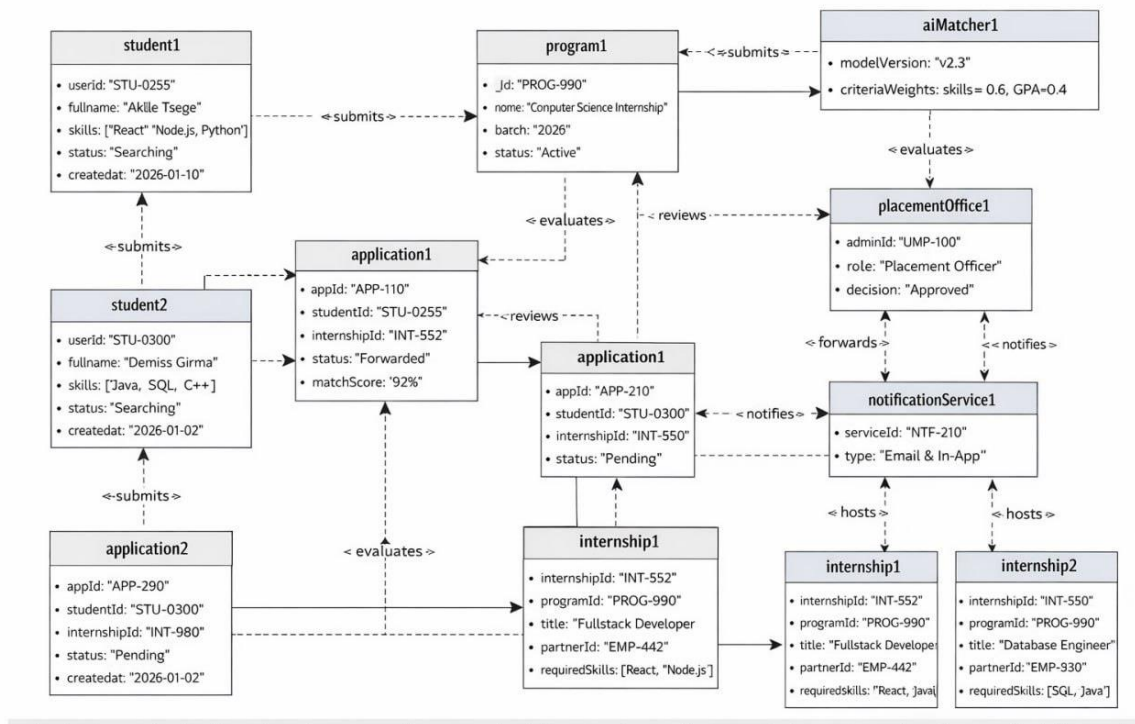


Figure 3-15: Object Model Diagram

3.1.6 DATA DICTIONARY

The Data Dictionary is a comprehensive reference that documents the database schema and data elements used in the Smart Internship Placement for University Students system. It provides a structured definition of each MongoDB collection, field name, data type, and constraint required to support the platform's core functionalities.

This section serves as a technical blueprint for the backend architecture, ensuring consistency in how critical data—such as persistent student academic profiles, internship role requirements, and AI-generated matching scores is stored and managed. By clearly defining these elements, the Data Dictionary facilitates the implementation of the Managed Brokerage workflow, ensuring that data integrity is maintained as information flows between Students, UIL Coordinators (Department Head), and Industry Partners.

Field Name	Data Type	Constraints	Description
User Id	String	PK, Unique	Unique identifier (e.g., "STU-0255" or "IP-442").
Full Name	String	Required	Full name of the student or Industry Partner rep.
email	String	Unique, Required	Registered email used for system login.
password	String	Encrypted	Bcrypt hashed password for security.
role	String	Required	User type: 'Student', 'Industry Partner', or 'Admin'.
Is Verified	Boolean	Default: False	Verification status (Critical for Industry Partners).
Created At	Date	Auto-generated	Timestamp when the account was registered.

Table 3-18: User Collection

Field Name	Data Type	Constraints	Description
Program Id	String	PK, Unique	Unique ID for the batch (e.g., "PROG-990").
name	String	Required	Title of the program (e.g., "Summer 2026 Internship").
Batch Year	String	Required	The academic year target (e.g., "2026").
status	String	Default: "Active"	Current state: 'Active' or 'Closed'.
description	String	Optional	General guidelines for the internship season.

Table 3-19: Program Collection

Field Name	Data Type	Constraints	Description
Internship Id	String	PK, Unique	Unique ID for the opportunity (e.g., "INT-552").
Program Id	String	FK	Links to the specific Program Batch.
Partner Id	String	FK	The ID of the Industry Partner (e.g., "IP-442").
title	String	Required	Role title (e.g., "Full Stack Developer").
Required Skills	Array	Required	List of skills (["React", "Node.js"]) for AI Matching.
stipend	String	Optional	Financial compensation details.

SMART INTERNSHIP PLACEMENT FOR UNIVERSITY STUDENTS

status	String	Default: "Pending"	Status: 'Pending Approval', 'Active', 'Closed'.
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Table 3-20: Internship Collection

Field Name	Data Type	Constraints	Description
App Id	String	PK, Unique	Unique application identifier (e.g., "APP-110").
Student Id	String	FK	Links to the Student applying.
Internship Id	String	FK	Links to the specific Internship opportunity.
Match Score	String	Calculated	NLP generates the AI Similarity Score (e.g., "92%").
status	String	Default: "Under Review"	Stage: 'Under Review', 'Forwarded', 'Accepted'.
Applied At	Date	Auto-generated	Timestamp of submission.

Table 3-21: Application Collection

Field Name	Data Type	Constraints	Description
Notification Id	String	PK, Unique	Unique ID for the alert message.
Recipient Id	String	FK	User ID of the person receiving the alert.
message	String	Required	Content (e.g., "Your Profile has been Forwarded").
Is Read	Boolean	Default: False	Tracks if the user has opened the notification.
Created At	Date	Auto-generated	The time when the

			event was triggered.
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Table 3-22: Notification Collection

3.1.7 CLASS MODEL

The Class Model illustrates the static architecture of the Smart Internship Placement System, defining the system's classes, attributes, and relationships. The design follows an Object-Oriented approach based on an inheritance structure, where the User class serves as the foundation for three distinct actors: Student, Industry Partner, and UIL Coordinator (Department Head). This ensures a unified authentication mechanism while allowing for role-specific attributes, such as GPA for Students and verified status for Industry Partners.

A central feature of this model is the Application class, which serves as the associative link between a Student and an Internship. Crucially, the Application class stores the match Score, which is dynamically generated by the system's AI Engine using NLP analysis. This score quantifies the alignment between the student's skills array and the internship's required skills. Furthermore, the model enforces the "Managed Brokerage" logic by explicitly connecting the UIL Coordinator (Department Head) to the application process, ensuring that they must "Review" and "Forward" applications before an Industry Partner can view the student's profile.

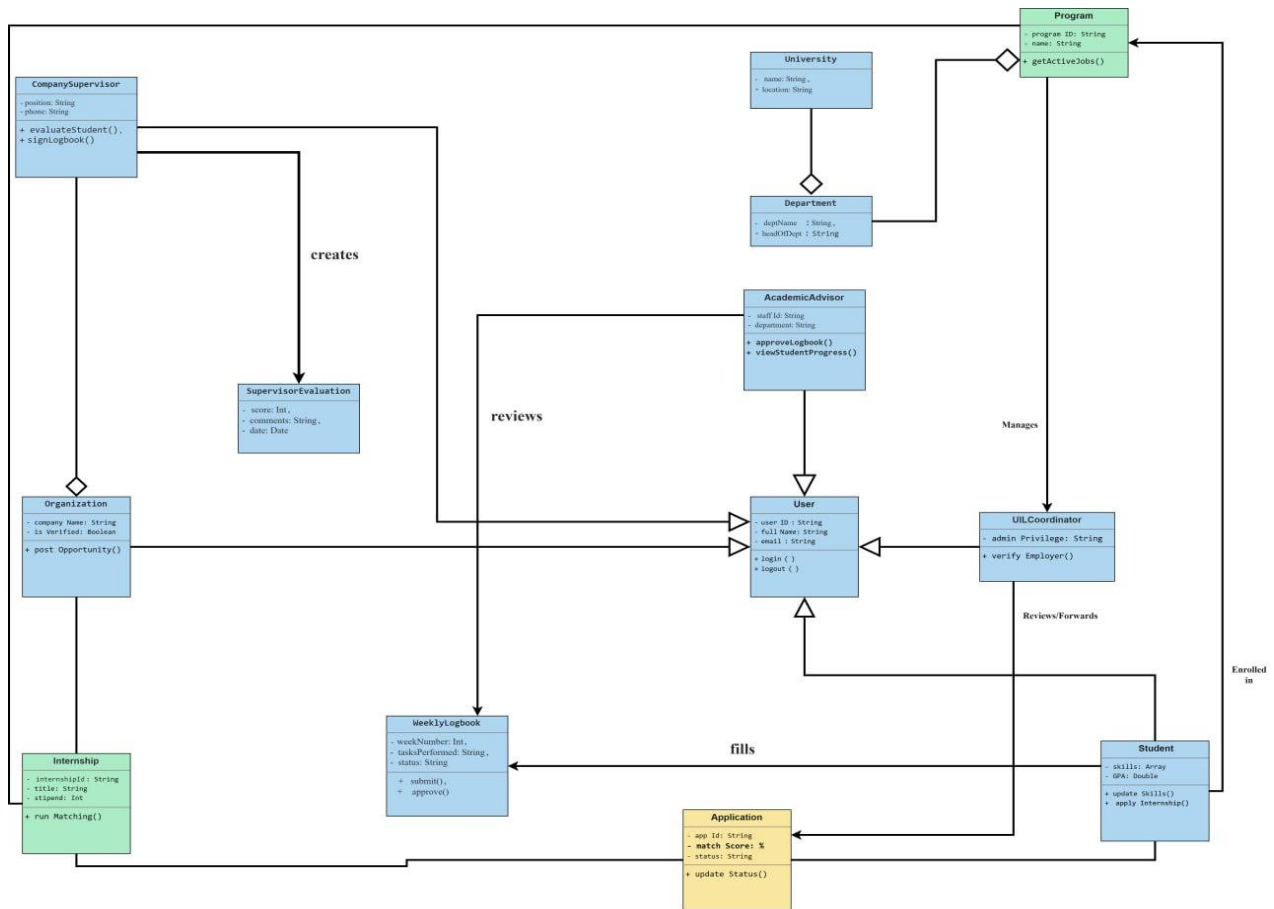


Figure 3-16: Class Model Diagram

3.1.8 DYNAMIC MODELING

Dynamic Modeling is a critical aspect of system design that captures and visualizes the runtime behavior of the Smart Internship Placement System. While static modeling defines the system's structure, dynamic modeling illustrates how the system responds to internal and external events over time. By mapping interactions, state transitions, and control flows, this section provides a comprehensive view of the system's operational logic. It specifically focuses on the secure authentication of distinct actors, the AI-driven matching execution, and the managed brokerage workflow, ensuring that the movement of data between Students, UIL Coordinators (Department Heads), and Industry Partners follows the strictly defined institutional business rules.

3.1.8.1 STATE CHART DIAGRAM

A State Chart Diagram describes the lifecycle of a specific object, illustrating the various states it transitions through in response to system events. In the context of this project, State Chart Diagrams are utilized to model two complex processes:

1. **User Authentication State:** Visualizes the journey of an actor (Student, Coordinator, or Partner) from the login request, through the JWT-based verification process, to the initialization of their role-specific dashboard or handling of authentication errors.
2. **Placement Application Lifecycle State:** Tracks a student's internship application as it progresses through the managed workflow. It details the transition from the initial 'Applied' state, to the 'Under Review' state (where the UIL Coordinator (Department Head) audits the AI Match Score), then to the 'Forwarded' state (visible to the Industry Partner), and finally to the 'Placement Confirmed' or 'Rejected' state based on the hiring decision.

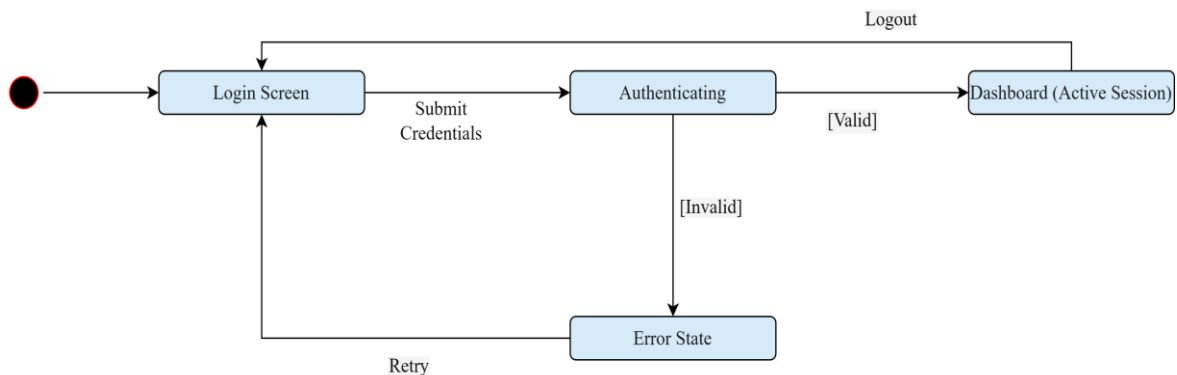


Figure 3-17: State Chart Diagram for User Login

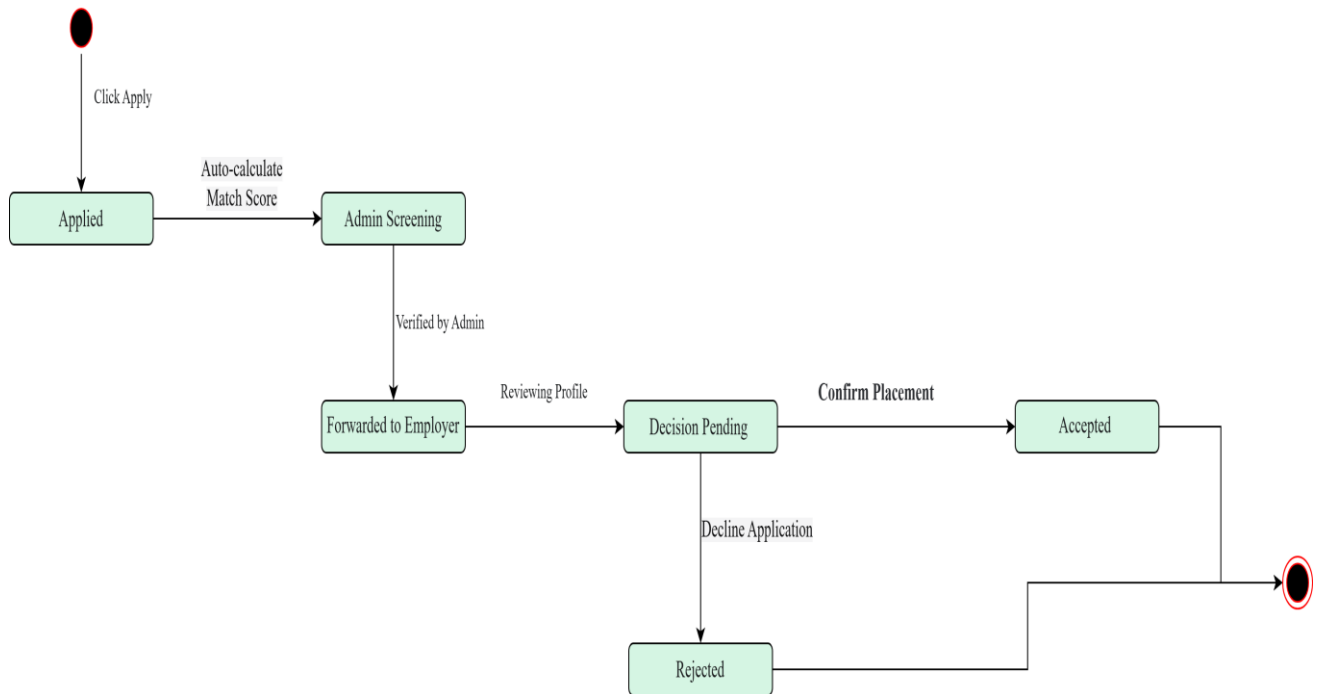


Figure 3-18: Internship Application Lifecycle State Chart

3.1.8.2 SEQUENCE DIAGRAM

A sequence diagram is a type of UML diagram that illustrates how objects interact with each other in a particular sequence of events. In the context of the Smart Internship Placement System, this diagram visualizes the high-speed interaction between the Student and the AI Matching Engine.

The flow begins when A Student requests matched opportunities through the React Frontend. The API Gateway coordinates the data retrieval from MongoDB, pulling both the student's Persistent Academic Profile and the active Internship Opportunities. The core logic is triggered when the AI Engine executes a Natural Language Processing (NLP) similarity analysis to calculate match scores based on technical competencies. Finally, the system ranks the results and returns a curated list of internships to the student, ensuring a highly relevant and curriculum-aligned placement process.

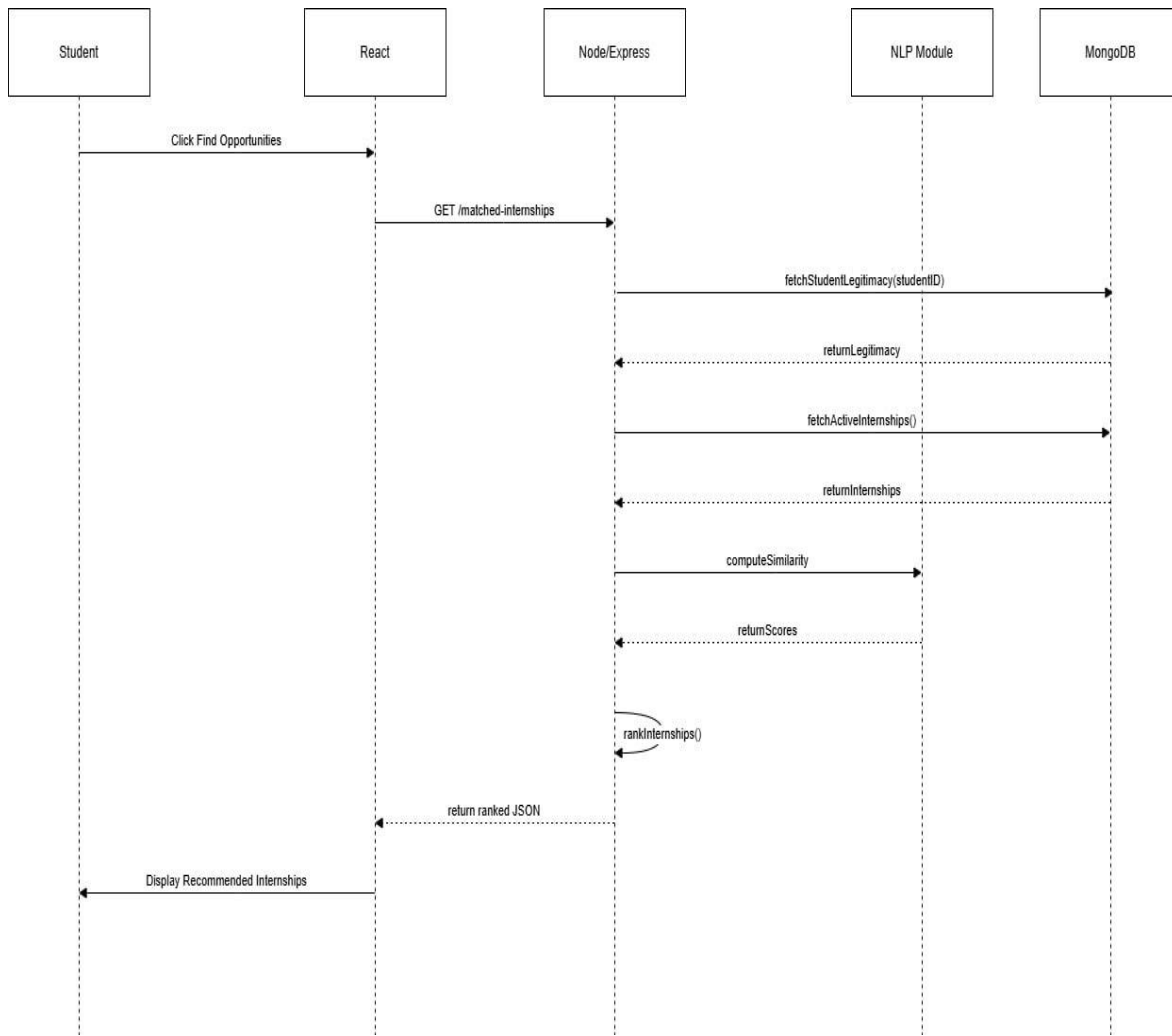


Figure 3-19: Sequence Diagram for AI Matching Process

Description of AI Smart Matching Sequence Diagram

The Sequence Diagram above provides a sophisticated visualization of the synchronous interactions between the Student actor and the system's backend components during the intelligent placement process. This workflow represents the architectural blueprint of the AI Matching Engine, ensuring that internship recommendations are mathematically derived rather than randomly assigned.

1. Request and Academic Data Aggregation:

- **Initiation:** The process commences when the Student interacts with the React Frontend to request personalized placement recommendations.

- **API Orchestration:** The frontend dispatches a secure request to the API Gateway (Node.js/Express). The gateway acts as the central conductor, ensuring that the request is authenticated before processing.
- **Dual-Retrieval Strategy:** The system executes two parallel queries to the MongoDB Database: first, it fetches the student's Persistent Academic Profile (specifically their technical skill vector), and second, it retrieves the repository of active Internship Opportunities posted by verified Industry Partners.

2. Algorithmic Processing (The AI Core):

- **NLP Injection:** Once the data is aggregated, the API Gateway forwards the raw skill sets and internship descriptions to the AI Engine (NLP Service).
- **Similarity Computation:** The engine triggers an internal Natural Language Processing (NLP) function. This utilizes the Vector Space Model (VSM) to execute a semantic analysis, comparing the student's academic competencies against the specific requirements of each internship.
- **Scoring:** The engine computes a precise Match Score (%) - a numerical value representing the degree of alignment between the student and the host organization.

3. Ranking and Deterministic Output:

- **Logic Execution:** The system applies a sorting algorithm (rank Jobs By Score) to organize the opportunities in descending order, prioritizing the highest relevance.
- **JSON Delivery:** The processed and ranked data is transmitted back to the frontend in a structured JSON format.
- **Final Presentation:** The student is presented with a curated list of Recommended Internships, enabling them to apply for roles where they have the highest probability of academic and professional success.

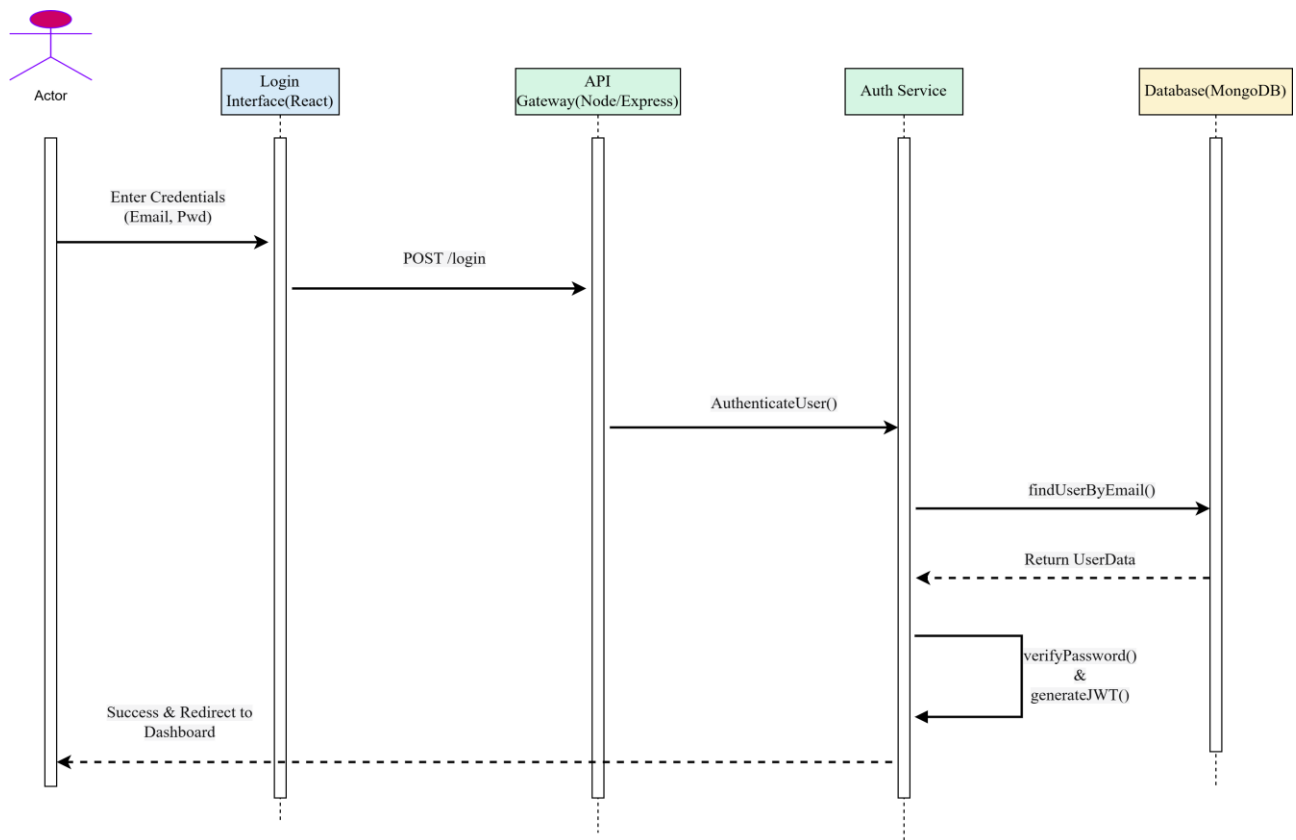


Figure 3-20: Sequence Diagram for Secure User Authentication

Description for User Login Sequence Diagram

The Sequence Diagram above details the secure authentication architecture within the Smart Internship Placement System. This process is critical for maintaining institutional data integrity by ensuring that Students, UIL Coordinators (Department Head), and Industry Partners are correctly identified and routed.

1. Credential Submission:

The process initiates when an actor (Student, Employer(Industry partner), or Admin) submits their credentials (Email and Password) via the React User Interface. The system enforces input validation at this stage to prevent malformed requests.

2. API Orchestration:

The API Gateway (Node.js/Express) receives the POST /login request. Acting as the system's security perimeter, it forwards the credentials to the specialized **Auth Service** for verification.

3. Cryptographic Verification & Data Lookup:

The Auth Service queries the MongoDB Database to retrieve the user's record. Upon locating the email, the system executes a security function (`verifyPassword()`) using Bcrypt to compare the hashed password. Simultaneously, it checks the user's Role (e.g., 'Student' vs. 'UIL Coordinator (Department Head)') to enforce Role-Based Access Control (RBAC).

4. Session Initialization:

Upon successful verification, the system generates a signed JSON Web Token (JWT). This token is returned to the client, establishing a secure, stateless session and automatically redirecting the user to their authorized Dashboard (e.g., the Coordinator Panel or Student Profile).

3.1.9 USER INTERFACE

The User Interface (UI) design describes how the system presents critical placement data to stakeholders and facilitates interaction with the AI core. The Smart Internship Placement System utilizes a responsive web-based interface built with React.js. The design philosophy prioritizes academic professionalism, intuitive navigation, and clear data visualization—specifically for the AI Matching Scores and the Managed Brokerage workflow.

1. Authentication Interface (Sign In)

The Sign-In interface serves as the secure, centralized gateway for all three institutional actors: Students, Industry Partners, and the UIL Coordinator (Department Head). As shown in Figure 3-21, the interface features:

- **Role-Agnostic Access:** A unified login portal that automatically routes users to their specific dashboard based on their credentials.
- **Security Features:** Encrypted password fields and strict input validation to prevent unauthorized access.
- **Session Management:** A "Keep me logged in" option that securely manages JWT token persistence for user convenience.
- **Account Recovery:** Quick access links for password resets and new account registration requests.

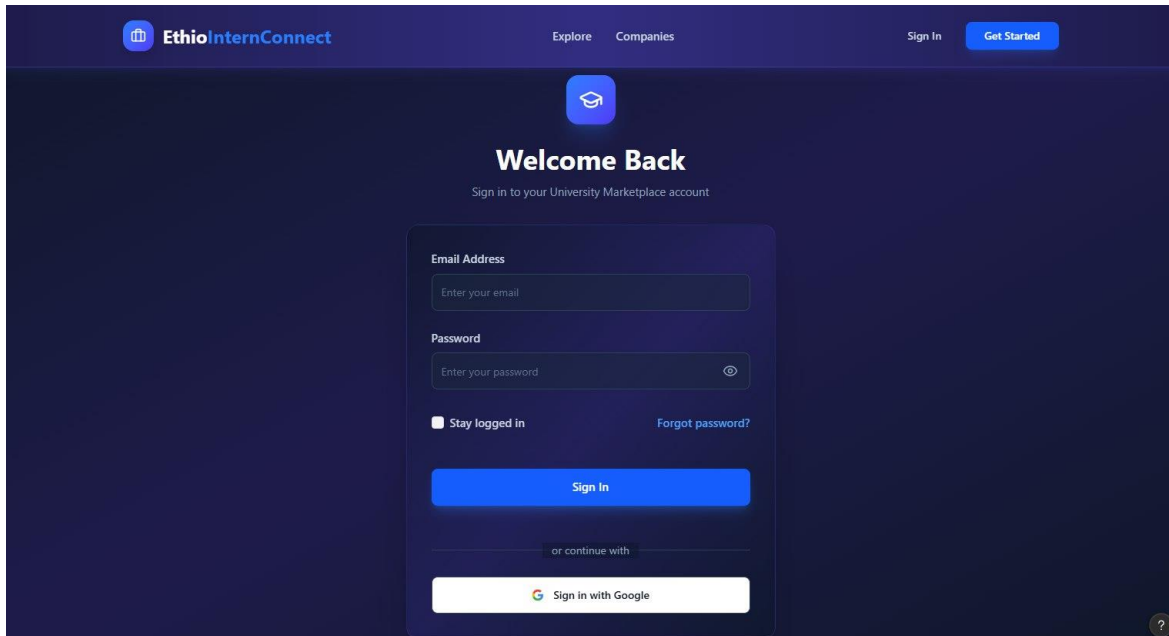


Figure 3.21 Login interface.

2. Student Dashboard (Smart Finder Interface)

The Student Dashboard is designed to be the central discovery hub for academic internships. It features:

- **AI Match Indicator:** A visual percentage-based tool that instantly shows how well the student's technical skills align with a specific internship requirement.
- **Internship Opportunity Feed:** A curated, scrollable list of curriculum-aligned internship roles, automatically prioritized and sorted by the AI Engine.
- **Real-Time Status Tracker:** An interactive timeline displaying the current stage of the application lifecycle (e.g., Applied -> Under Admin Review -> Forwarded to Industry -> Accepted).
- **Persistent Profile Editor:** A dedicated section where students maintain their digital academic identity by updating technical skills (tags), uploading CVs, and linking project portfolios (GitHub), which serves as the source data for the NLP matching engine.

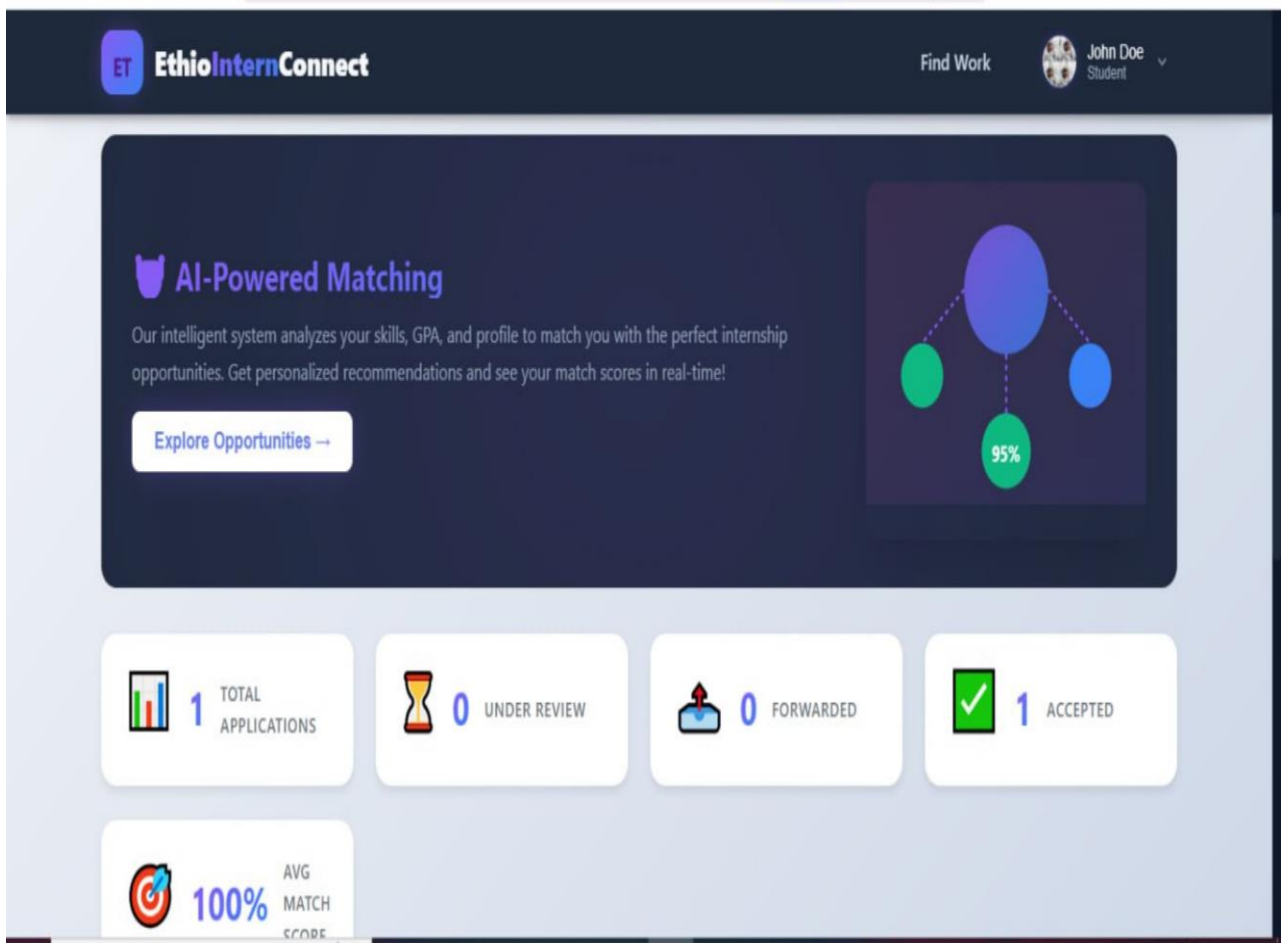


Figure 3.22 Student Dashboard

3. Admin Command Center (Brokerage Interface)

The Admin interface is built for the UIL Coordinator (Department Head) to maintain institutional oversight and exercise Managed Brokerage control:

- Partner Verification Panel: A dedicated table for auditing and approving new Industry Partner registration requests before they can post vacancies.
- AI Matching Analytics: A statistical view showing the distribution of skill matching scores across different departments.
- Managed Forwarding Console: A critical workspace where the Coordinator reviews top-ranked student profiles and executes the "Forward to Employer" action, ensuring that only qualified candidates are presented to the industry.
- Placement Reporting System: A feature that generates downloadable reports on successful linkages, identifying which departments have the highest placement rates and which industry partners are most active.

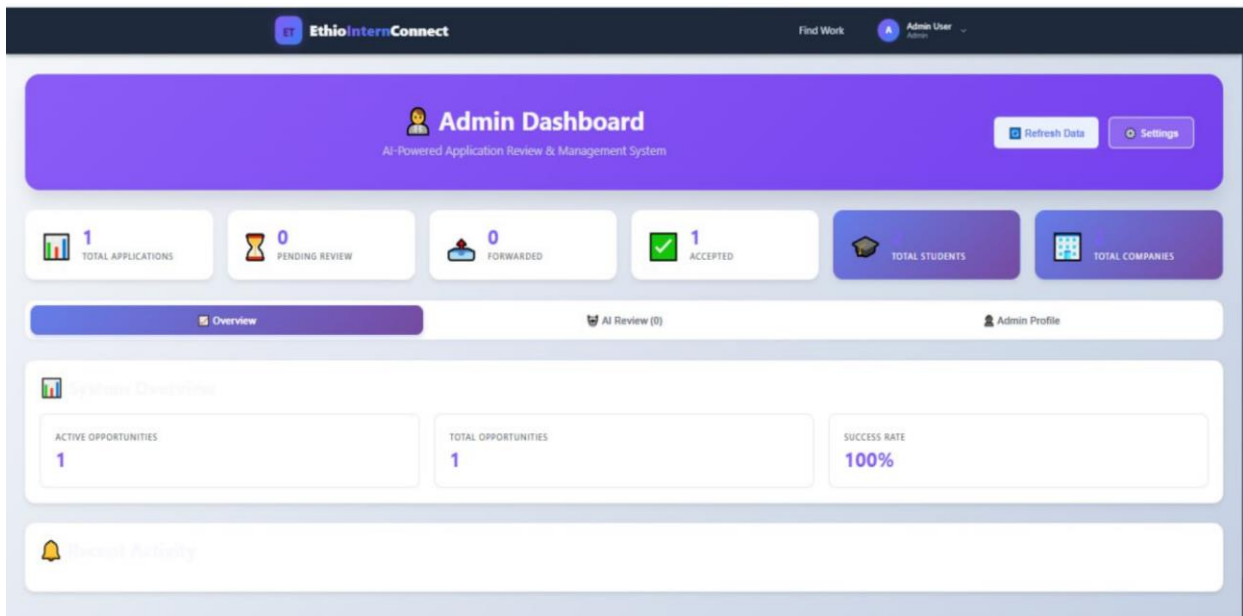


Figure 3.23 Admin Dashboard

4. Industry Partner Dashboard (Recruitment Interface)

The Industry Partner interface is streamlined to focus on opportunity creation and candidate evaluation without administrative clutter. It features:

- **Vacancy Management Console:** A tool for drafting, editing, and posting internship requirements. It includes mandatory fields for "Required Technical Skills" to assist the AI vectorization process.
- **Vetted Candidate Pool:** A restricted view where employers can only see student profiles that have been strictly verified and forwarded by the UIL Coordinator (Department Head), ensuring quality control.
- **Feedback Loop Module:** A decision-making interface allowing employers to mark candidates as "Accepted" or "Rejected" with mandatory feedback comments, which are routed back to the university for record-keeping.

SMART INTERNSHIP PLACEMENT FOR UNIVERSITY STUDENTS

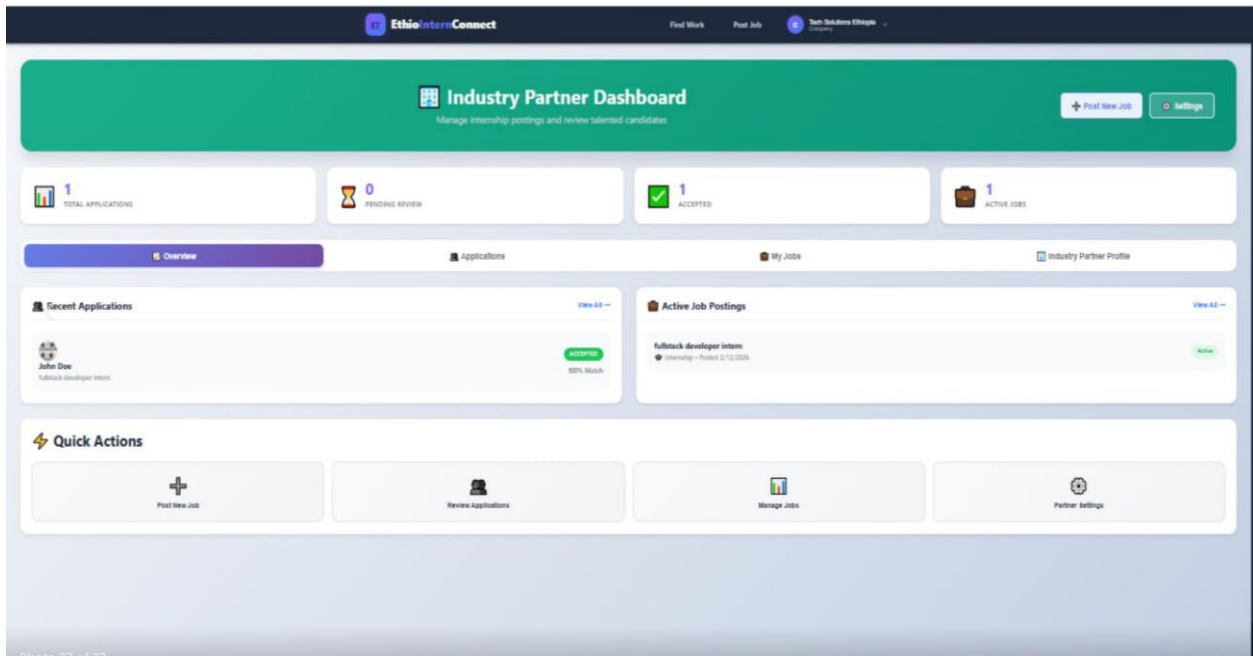


Figure 3.24 Industry Partner Dashboard

CHAPTER FOUR

SYSTEM DESIGN

4.1 INTRODUCTION

This chapter outlines the architectural design and software principles used to develop the Smart Internship Placement System. Our primary goal was to replace the manual, paper-based workflow at the University of Gondar with a centralized web-based platform. We prioritized a "Managed Brokerage" model, meaning the system doesn't just randomly connect students and employers; it allows the UIL (Department) office to oversee the process. Additionally, we integrated an AI matching component using Natural Language Processing (NLP) to mathematically calculate how well a student's skills fit a specific internship requirement.

The design focuses on four main goals:

Modularity: We divided the system into independent parts—Authentication, Matching Engine, and Management—so that if one part fails or needs an update, it doesn't break the whole system.

Performance: We used Node.js for the backend to handle multiple requests efficiently, while React.js ensures the user interface loads quickly on both phones and computers.

Data Security: Since we are handling student grades and personal info, we implemented role-based access so that only authorized coordinators and verified employers can see sensitive data.

Data Consistency: We used MongoDB to safely store student profiles and internship records, ensuring the university has a permanent digital archive.

4.2 CURRENT SOFTWARE ARCHITECTURE

Currently, the University of Gondar manages internships through a decentralized, manual system. There is no dedicated software architecture in place. Through our analysis, we identified two main environments:

The Internal Manual Workflow

The UIL office currently relies on physical files and excel sheets. Students submit hard copies of CVs, which are stored in cabinets. This creates a bottleneck because retrieving a specific student's record takes time. Furthermore, the "matching" process is done manually

by coordinators. Without a tool to analyze skills, coordinators often assign students based on availability rather than technical fit, leading to mismatches.

External Platforms

While students sometimes use sites like LinkedIn, these platforms are not suitable for academic internships. They lack the university's oversight (brokerage) layer. The university cannot verify if the company is legitimate, nor can it ensure the work aligns with the computer science curriculum.

Therefore, the proposed Smart Internship Placement System acts as a bridge, bringing the efficiency of digital platforms while maintaining the university's necessary control.

4.3 PROPOSED SOFTWARE ARCHITECTURE

To solve the problems mentioned above, we designed a Three-Tier Architecture using the MERN Stack. This separates the user interface, the logic, and the database.

Presentation Layer (Client): Built with React.js. This is what the students and coordinators see. It handles the data visualization and sends requests to the server.

Application Logic Layer (Server): Built with Node.js and Express. This is the brain of the system. It processes the API requests, runs the brokerage rules (checking if a student is eligible), and executes the Python-based NLP matching logic.

Data Persistence Layer (Database): Built with MongoDB. We chose this NoSQL database because student profiles and internship descriptions vary in structure, and MongoDB handles this flexibility well.

4.3.1 System Decomposition

Instead of building one massive block of code, we decomposed the system into four manageable subsystems.

1. Identity and Access Management (IAM)

This subsystem handles security. Since we cannot allow unauthorized access, we implemented a "Zero Trust" approach.

Authentication: We use JSON Web Tokens (JWT). When a user logs in, they get a secure token. This is more scalable than traditional sessions. Passwords are never stored as plain text; we hash them using Bcrypt to protect against hacks.

Role-Based Access Control (RBAC): The system strictly separates what users can do. Students can only edit their profiles and apply. Industry partners can post jobs but only see

students forwarded to them. The UIL Coordinator (Department Head) has full control to verify users and manage the process.

2. Managed Brokerage & Opportunity Subsystem

This is the operational core that digitizes the UIL office's work.

Opportunity Management: When a company posts an internship, it doesn't go live immediately. It goes into a "Pending Approval" queue. The UIL coordinator (Department Head) checks if the job meets academic standards before students can see it.

Workflow Logic: This handles the application flow. When a student applies, the application goes to the Coordinator's Console, not the employer. The coordinator reviews it first. The system tracks the status (Applied → Under Review → Forwarded → Accepted/Rejected) so everyone knows where the process stands.

3. Intelligent Computation Subsystem (The AI Engine)

This is the technical highlight of the project. It automates the screening process.

Vectorization: We use an NLP library to clean up the text from student CVs and job descriptions (removing words like "and", "the"). It then converts the skills into numerical vectors using TF-IDF.

Matching: The system calculates the Cosine Similarity between the student's vector and the job's vector. If the score is high (e.g., >80%), it suggests a strong match. This helps coordinators make data-driven decisions rather than guessing.

4. Presentation and Interface Subsystem

This layer ensures the system is usable.

User Interface: We used React components to create a modular design. For example, the "Job Card" component is reused across different pages.

API Gateway: The Express router acts as a traffic controller. It receives requests from the frontend, validates the data to prevent attacks (like SQL Injection), and routes it to the correct service.

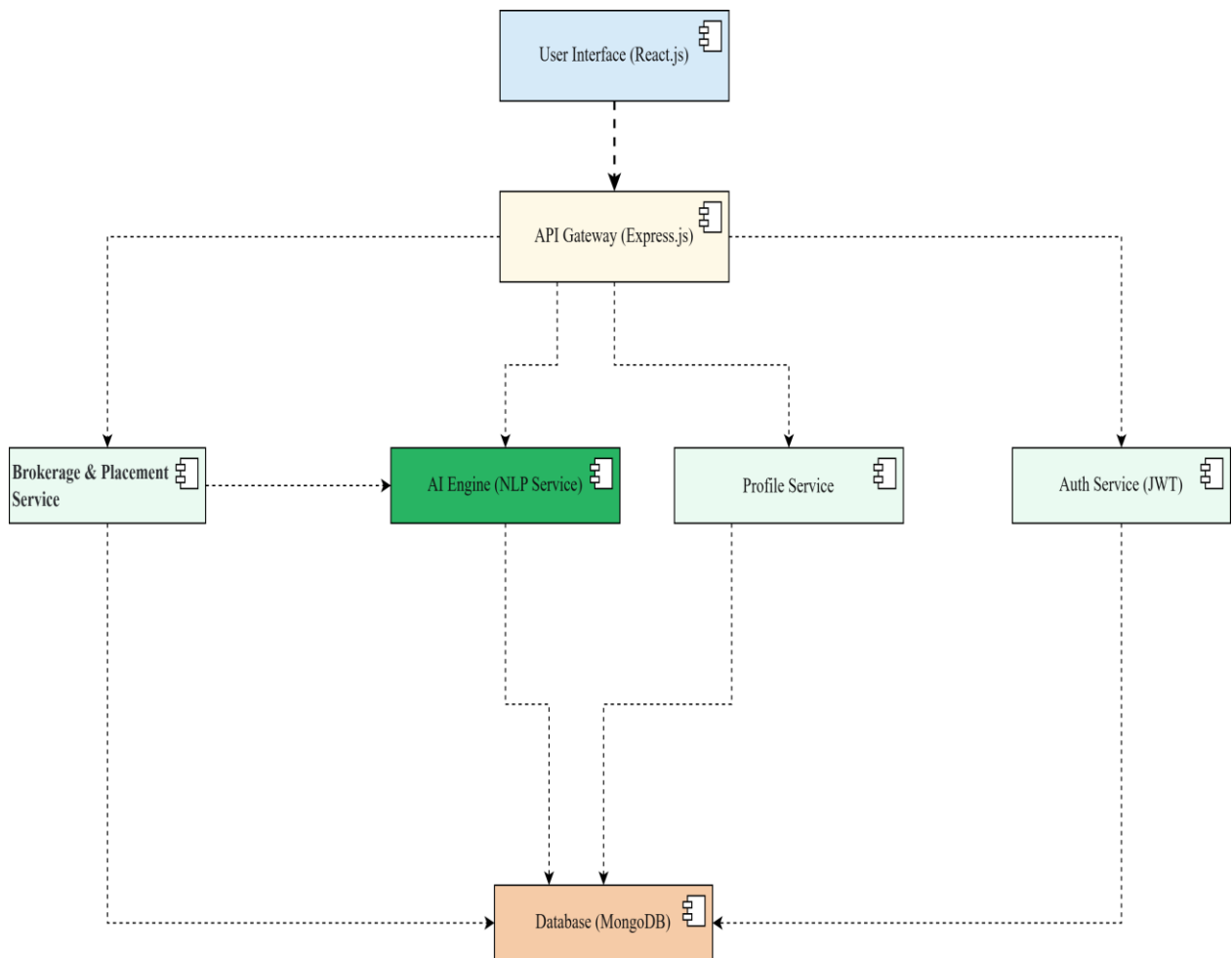


Figure 4-11: Component Diagram

4.3.2 HARDWARE/SOFTWARE MAPPING

The Diagram illustrates how the Smart Internship Placement System is distributed across different hardware nodes.

1. **Client Node:** Students, Industry Partners, and UIL Coordinators (Department Head) access the system via standard web browsers, where the React.js frontend is rendered.
2. **Application Server:** This central node hosts the Node.js/Express runtime, the API Gateway, the Managed Brokerage Service, and the specialized AI Engine, which handles the heavy lifting of NLP analysis.
3. **Database Node:** A dedicated server manages the MongoDB NoSQL storage to ensure data persistence and high availability. Communication between these nodes is secured using the HTTPS protocol for user interactions and optimized database protocols for internal backend services.

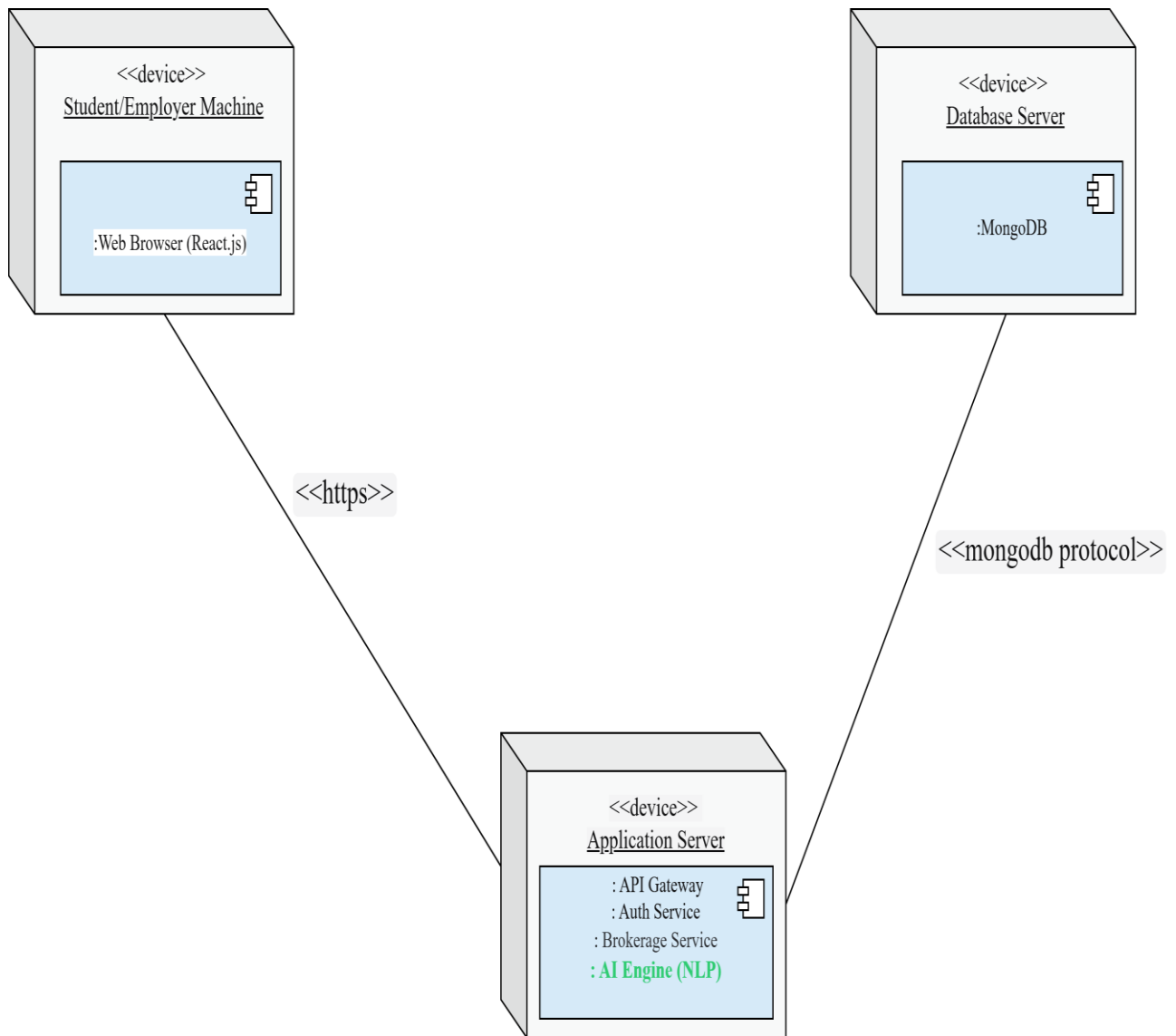


Figure 4-12: Hardware/Software Mapping Diagram

4.3.3 PERSISTENT DATA MODELING

Persistent data modeling focuses on the architectural design of how the system captures, stores, and manages data to ensure its availability, consistency, and integrity. Unlike transient session data, persistent data in this system—such as academic records, AI-generated match scores, and internship placement histories—must be preserved securely to support long-term career tracking. This section defines the database selection strategy,

schema definitions, and the relational mapping required to support the AI-powered Matching Engine and the Managed Brokerage Workflow.

1. Database Selection

For the Smart Internship Placement System, the NoSQL (MongoDB) database was selected as the primary storage solution. This choice is strategically aligned with the MERN stack and the specific requirements of the AI algorithm:

- **NoSQL Database (MongoDB):**
 - ❖ **Schema Flexibility:** Chosen to handle semi-structured data, particularly the variable nature of student skill sets (arrays) and unorganized internship descriptions. This allows for seamless integration with the Natural Language Processing (NLP) engine without complex schema migrations.
 - ❖ **Scalability:** Supports horizontal scaling (Sharding) to efficiently manage thousands of concurrent student profiles and growing archival data as the university expands the program.
 - ❖ **High Performance:** Facilitates faster retrieval of nested documents—such as a student’s project history embedded within their profile—reducing read latency during the real-time matching process.

2. Database Schema & Tables

The database architecture is organized into logical collections (tables) that govern the placement ecosystem. These collections ensure data integrity while enforcing the "Managed Brokerage" business rules.

A. User Management Tables

These tables handle system access, security, and identity verification.

1. User Collection

- ❖ **Purpose:** Acts as the centralized authentication layer.
- ❖ **Function:** Stores core credentials (Email, Bcrypt-hashed Password) and system-wide identifiers.
- ❖ **Role Management:** Includes a role field to strictly differentiate between Students, Industry Partners (Employers), and University Coordinators (Admins), enforcing Role-Based Access Control (RBAC).

2. Student Profile Collection

- ❖ **Purpose:** Represents the Persistent Academic Identity of the student.
- ❖ **Function:** Stores verifiable professional data such as Department, GPA, and uploaded CVs.
- ❖ **AI Integration:** Includes a dedicated Technical Skills array (e.g., ["React", "Data Analysis"]), which serves as the primary input vector for the TF-IDF matching algorithm.

3. Employer (Industry Partner) Collection

- ❖ **Purpose:** Manages the institutional profiles of participating companies.
- ❖ **Function:** Stores company details, industry sectors, and contact information.
- ❖ **Trust-Mechanism:** Includes Verification Status field (Pending vs Verified). This ensures that only legitimate organizations vetted by the University Coordinator can post internship opportunities.

B. Placement & Brokerage Tables

These tables digitize the internship workflow, replacing physical files with a structured digital pipeline.

1. Program (Batch) Collection

- ❖ **Purpose:** Manages the academic timeline and internship cycles.
- ❖ **Function:** Stores batch information (e.g., "Graduating Class of 2026") and strict deadlines.
- ❖ **Linkage:** Acts as a container for opportunities, ensuring that internships are only posted for active and valid academic sessions.

2. Opportunity (Internship) Collection

- ❖ **Purpose:** Represents the specific vacancy or role offered by an employer.
- ❖ **Function:** Stores detailed job descriptions, stipend transparency data, and duration.

- ❖ **AI Target:** Contains the Required Skills and Job Description text fields, which the AI engine analyzes to calculate similarity scores against student profiles.

3. Application Collection (The Core Logic)

- ❖ **Purpose:** Acts as the associative entity connecting a Student to an Opportunity.
- ❖ **AI Intelligence:** Stores the mathematically calculated Match Score (%), allowing coordinators to rank candidates based on merit.
- ❖ **Workflow Tracking:** Tracks the application lifecycle through distinct states: Applied → Under Review (Coordinator Audit) → Forwarded (Sent to Employer) → Accepted or Rejected.

C. System Interaction Tables

These tables facilitate communication and audit trails.

1. Notification Collection

- ❖ **Purpose:** Manages asynchronous communication and real-time alerts.
- ❖ **Function:** Stores system-generated messages regarding status updates (e.g., "Your profile has been forwarded"), ensuring users are instantly informed of critical changes without needing to refresh the dashboard manually.

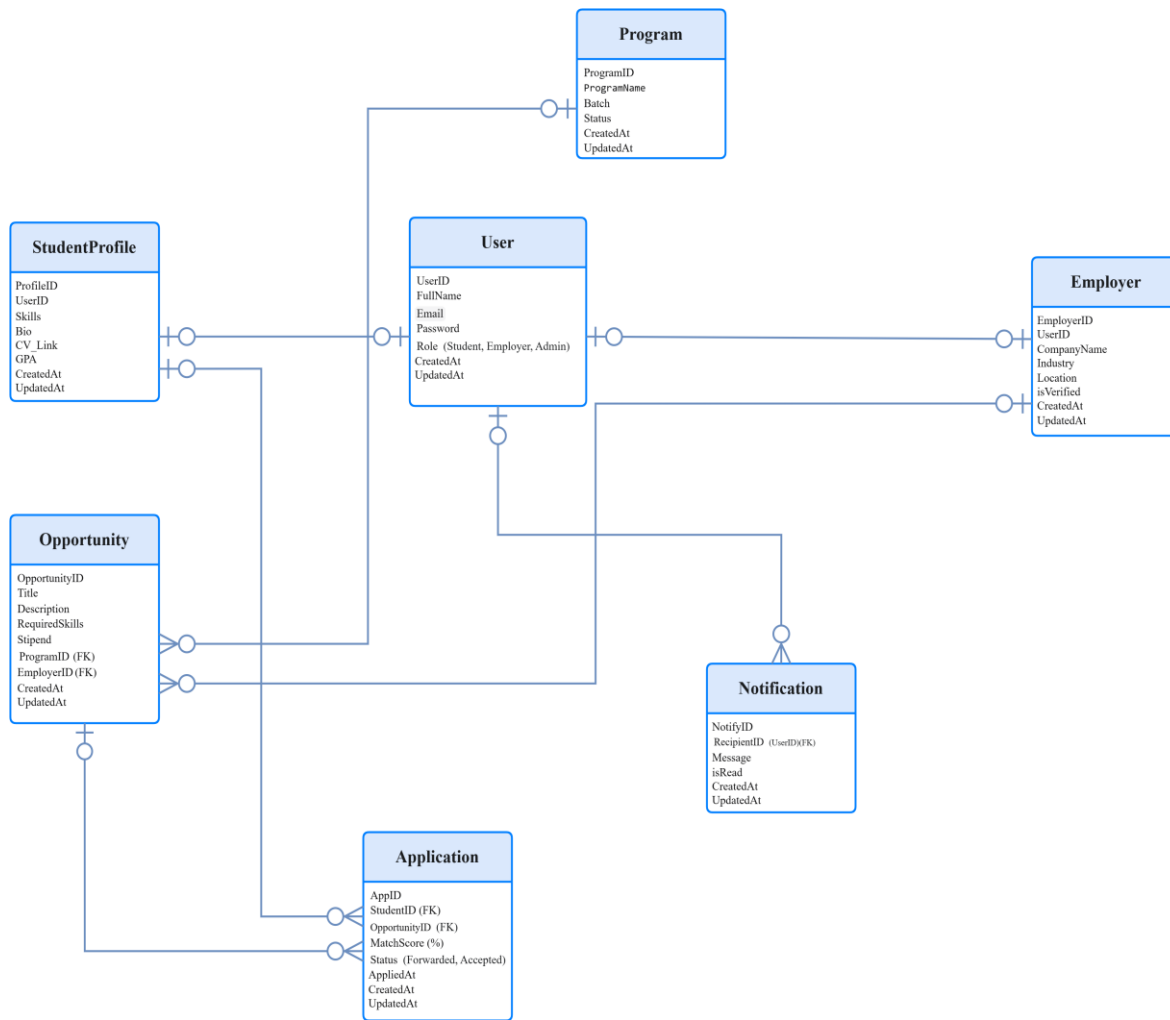


Figure 4-13: Persistence data diagram

Description of Relations in the ER Diagram:

The Entity-Relationship (ER) Diagram illustrates the logical structure of the database with the following key associations:

- ❖ **User to Profiles:** A one-to-one (1:1) relationship exists between User and Student Profile, as well as User and Employer. This segregates authentication credentials from professional data.
- ❖ **Program to Opportunity:** A one-to-many (1:M) relationship ensures that Opportunities are strictly linked to a specific

academic Program (Batch), allowing the university to manage internship seasons effectively.

- ❖ **The Brokerage Logic (Application):** The Application entity acts as a junction table between Student Profile and Opportunity. It stores the Match Score (%), which represents the AI-calculated compatibility, and the **Status** field, which tracks the application's lifecycle (e.g., Forwarded, Accepted).
- ❖ **Verification Flow:** The Employer entity includes an **is Verified** attribute, enforcing the institutional rule that unverified partners cannot post opportunities or receive student applications.

4.3.4 ACCESS CONTROL AND SECURITY

The system implements a rigorous Role-Based Access Control (RBAC) mechanism designed to maintain the integrity of the University-Industry Linkage (UIL) process. By enforcing strict separation of privileges, the architecture ensures that users interact only with features relevant to their professional responsibilities, protecting sensitive student academic data from unauthorized access.

Role	Privileges and Access Rights
University Admin (Broker)	Central Oversight: Verify and approve Industry Partner accounts, audit all internship postings for curriculum alignment, and manage system-wide logs. Brokerage Execution: Review AI-generated match scores, perform the exclusive "Forward" action to share pre-screened student profiles with companies, and publish final placement results.
Employer (Industry Partner)	Opportunity Management: Register institutional profiles and post detailed

	internship opportunities, including stipend and skill requirements. Candidate Review: View <i>only</i> the specific student profiles that have been vetted and forwarded by the University Admin. Provide formal hiring decisions (Approve/Reject) to close the feedback loop.
Student	Profile Management: Create and update a Persistent Academic Profile (Skills, Bio, GitHub Links, and CV uploads). AI Discovery: Use the "Smart Finder" to browse AI-ranked internship opportunities, submit applications for specific roles, and track real-time application status (e.g., <i>Under Review</i> vs. <i>Forwarded</i>).

Table 4-11: Role-Based Access Control (RBAC) Specifications

Security Implementation Measures

To ensure a secure, trusted, and resilient digital environment, the system incorporates the following industry-standard security protocols:

- **JWT-Based Stateless Authentication:**

The system utilizes JSON Web Tokens (JWT) for secure session management. Upon successful login, the server issues a signed token that validates the user's identity for every subsequent request. This ensures that only authenticated users can access their specific dashboards (Student, Employer, or Admin) without burdening the server with session storage.

- **Cryptographic Password Hashing:**

User credentials are never stored in plain text. The system employs the Bcrypt hashing algorithm (with 10 salt rounds) to encrypt passwords before they are committed to the MongoDB database. This protects user accounts against brute-force attacks and rainbow table exploits.

- **Privacy-Preserving Brokerage Logic:**

A core feature of the system's architecture is "Data Privacy by Design." Student contact details (Phone numbers and Email addresses) remain hidden from Employers by default. These details are only revealed programmatically after the University Admin explicitly performs the "Forward" action, ensuring that students are protected from unsolicited contact or unverified organizations.

- **End-to-End Encryption (SSL/TLS):**

All data transmission between the React.js client and the Node.js server is encrypted using SSL/TLS protocols (HTTPS). This prevents "Man-in-the-Middle" (MitM) attacks and ensures that sensitive academic data and internship contracts cannot be intercepted during transit.

- **Robust Input Validation & Sanitization:**

To prevent injection attacks, the system validates all incoming data streams.

- ❖ **NoSQL Injection Prevention:** API requests are sanitized to prevent malicious queries from manipulating the MongoDB database.
- ❖ **XSS Protection:** Inputs for internship descriptions and student bios are escaped to prevent Cross-Site Scripting (XSS) attacks.

4.3.5 DETAILED CLASS DIAGRAM

The detailed class diagram illustrates the object-oriented architectural design of the Smart Internship Placement System. It defines the attributes (data) and methods (logic) for each component, enforcing a strict separation of concerns. The design utilizes an inheritance hierarchy where the User class serves as the base for Student, Employer (industry partner), and UIL Coordinator (Department Head), ensuring a unified authentication mechanism across all actors.

A central feature of this model is the AI Engine class, which encapsulates the Natural Language Processing (NLP) logic to calculate the match Score by comparing student skills against specific Internship requirements. The UIL Coordinator (Department Head) class acts as the system's central authority, enforcing the "Managed Brokerage" workflow. This allows the coordinator to verify Employer (industry partner) accounts, manage

academic Programs (Batches), and control the placement process through the forward Application() method, ensuring that only qualified and pre-screened students are connected with industry partners.

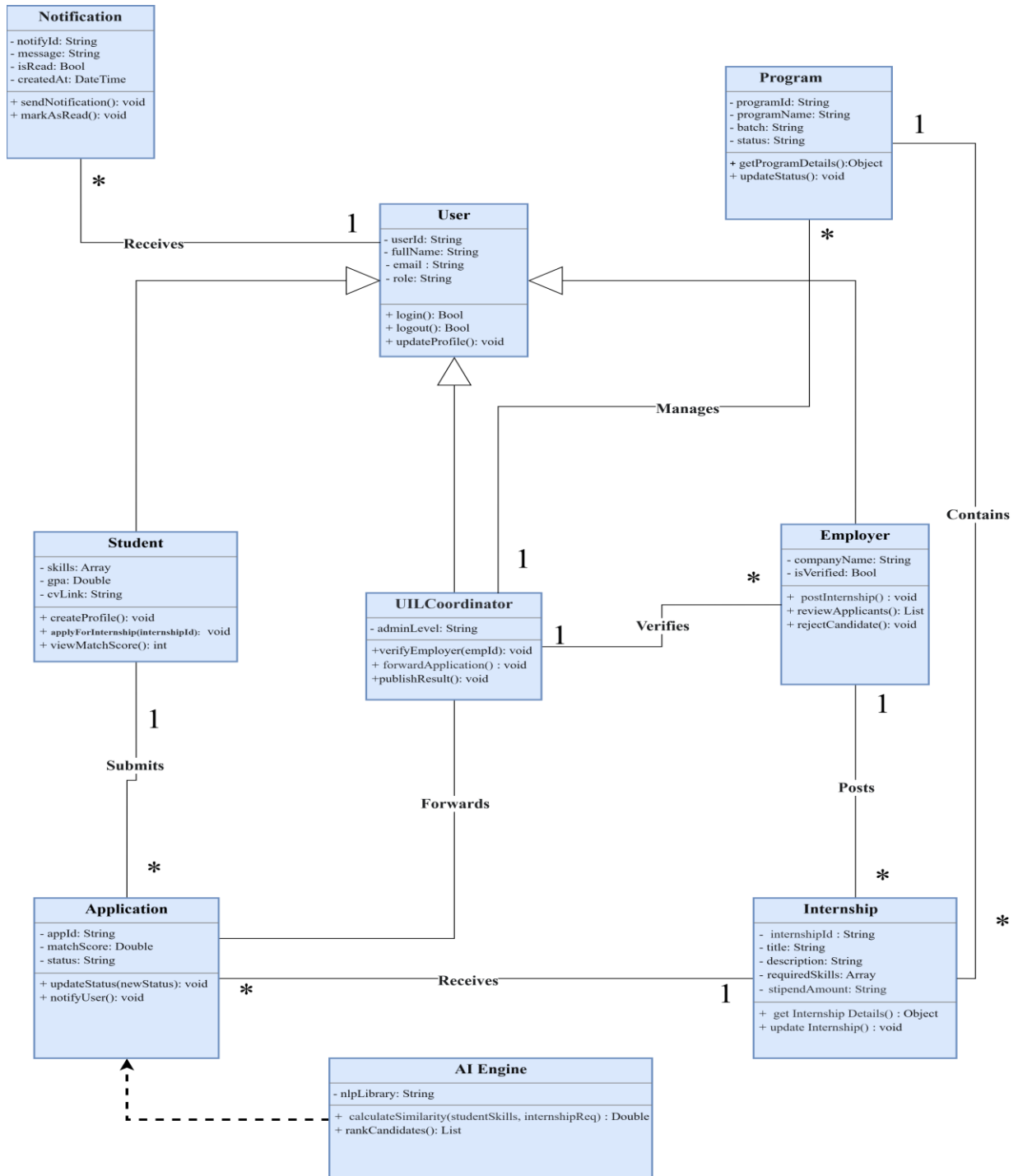


Figure 4-14: Detailed Class Diagram

4.3.6 PACKAGE DIAGRAM

The system's codebase is organized into modular packages to ensure separation of concerns, scalability, and maintainability. Each package encapsulates specific functionalities of the Smart Internship Placement System, communicating through well-defined interfaces as shown in Figure 4-15.

- **Identity Access Subsystem:** Handles secure user login, registration, and JWT (JSON Web Token) verification for Students, Employers, and UIL Coordinators (Department Head). It enforces Role-Based Access Control (RBAC) protocols via the Auth Service.
- **ClientPresentationLayer:** Represents the user-facing components, including the User Interface (React.js) and the API Gateway, which acts as the secure entry point, routing client requests to the appropriate backend microservices.
- **Internship Management Subsystem:**
 - ❖ **Internship Service:** Manages the lifecycle of internship programs, curriculum-aligned vacancy postings, and the validation of opportunity requirements.
 - ❖ **Placement Brokerage Service:** The core operational engine that manages student applications and enforces the "Managed Brokerage" workflow, ensuring applications are reviewed by coordinators before reaching industry partners.
- **Intelligent Matching Subsystem:**
 - ❖ **NLP Engine:** Processes textual descriptions using Natural Language Processing to extract key technical skills.
 - ❖ **ScoringEngine:** Implements the TF-IDF and Cosine Similarity algorithms to calculate a precise Match Score (%) between a student's profile and an internship vacancy.

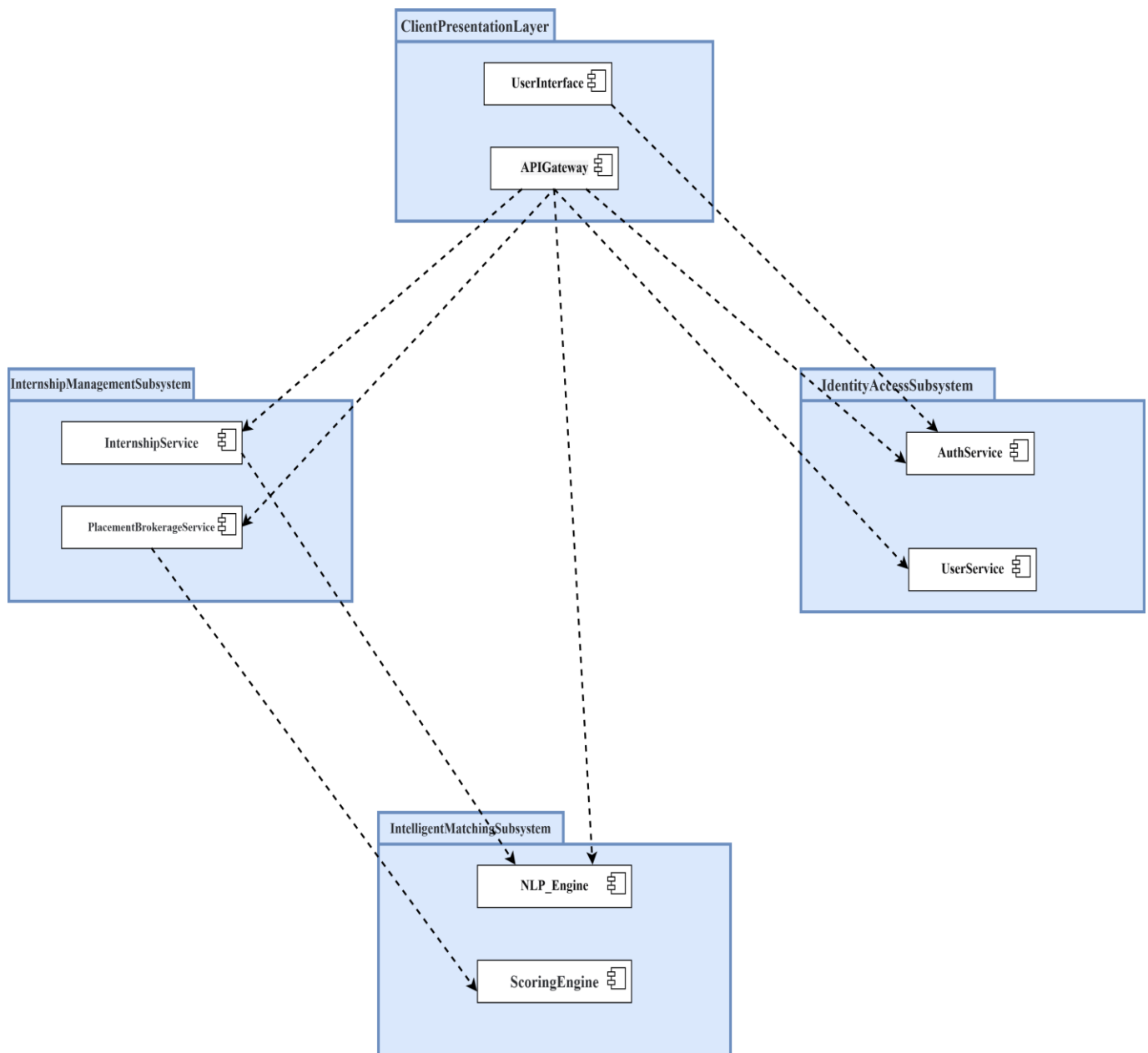


Figure 4-15: UML Package Diagram

4.3.7 DEPLOYMENT

Deployment represents the final and most critical phase of the system development life cycle (SDLC). It involves transitioning the Smart Internship Placement System from a local development environment into a secure, high-availability production ecosystem accessible to stakeholders over the web. This section details the comprehensive strategy for

deploying the platform, ensuring seamless, concurrent access for Students, Industry Partners, and University-Industry Linkage (Department Head) Coordinators.

The deployment architecture is designed to support the "Managed Brokerage" workflow, guaranteeing that the University Admin retains full operational control over the system while the AI Engine processes matching algorithms in real-time.

1. Deployment Architecture and Approach

The system adopts a Cloud-Based Microservices Architecture to ensure scalability, reliability, and 24/7 uptime. Unlike traditional monolithic deployments, this strategy decouples the client-side presentation layer from the server-side computational logic.

- **Hybrid Cloud Model:** The application logic is hosted on a secure cloud server to handle heavy AI processing, while the core database leverages a managed cloud service (Database-as-a-Service) to guarantee data persistence and automatic recovery.
- **Separation of Concerns:** The architecture enforces a strict separation where the React.js frontend is served as a static asset bundle, communicating with the Node.js/Express backend exclusively through secure RESTful APIs. This allows the frontend to load instantly on student devices while the backend handles the complex TF-IDF vectorization and brokerage logic independently.

2. Deployment Infrastructure Components

The production environment is built upon four robust pillars: The Server Runtime, the Client Interface, the AI Processing Unit, and the Data Persistence Layer.

A. Server-Side Environment (The Backend Core)

- ❖ **Operating System:** The backend is deployed on a Linux-based (Ubuntu LTS) cloud instance (e.g., AWS EC2, Digital Ocean, or Render). Linux is selected for its superior stability, security features, and efficient resource management.
- ❖ **Process Management with PM2:** To ensure high availability, the PM2 (Process Manager 2) production process manager is utilized. PM2 enables the Node.js application to run in Cluster Mode, utilizing all available CPU cores for faster AI processing. It also handles Automatic Restarts, ensuring

that if the system encounters a runtime error, it instantly reboots without downtime.

- ❖ **Reverse Proxy (Nginx):** An Nginx web server is configured in front of the Node.js application. It acts as a reverse proxy to handle load balancing, cache static content, and terminate SSL connections, adding an extra layer of security and performance.

B. Client-Side Hosting (The Presentation Layer)

- ❖ **Static Build Optimization:** The React.js source code undergoes a "Production Build" process. This involves Minification (removing unnecessary characters), Code Splitting, and Tree Shaking to reduce the bundle size significantly.
- ❖ **Content Delivery Network (CDN):** The optimized frontend files are deployed to a high-performance CDN (such as Vercel or Netlify). This ensures that students accessing the "Smart Finder" from different locations experience low-latency loading times, regardless of network speed.

C. Data Persistence Layer (Database)

- ❖ **Managed MongoDB Atlas:** The system utilizes MongoDB Atlas as a cloud-hosted database solution. It is configured with Replica Sets (redundant copies of data) to prevent data loss in case of server failure.
- ❖ **IP Whitelisting:** For enhanced security, the database cluster is configured to accept connections *only* from the specific IP address of the Backend Application Server, rejecting any unauthorized external access attempts.

3. Deployment Implementation Process

The transition to production follows a rigorous, multi-stage protocol to ensure security and functionality:

Stage 1: Environment Configuration & Security Hardening

- ❖ **Environment Variables:** Sensitive credentials—such as the DATABASE_URI, JWT_SECRET_KEY, and ADMIN_ACCESS_TOKENS—are migrated from local .env files to secure, encrypted server-side environment variables.

- ❖ **CORS Configuration:** Cross-Origin Resource Sharing (CORS) policies are strictly defined. The backend API is configured to allow requests *only* from the university's authorized frontend domain, preventing cross-site attacks.
- ❖ **SSL/TLS Encryption:** A valid SSL Certificate (e.g., via Let's Encrypt) is installed to enforce HTTPS. This ensures that all data transmitted between the student's browser and the server—including CVs, match scores, and stipend details—is encrypted in transit.

Stage 2: Build and Launch Sequence

- ❖ **Frontend Deployment:** The React application is built, optimized, and pushed to the hosting provider. The new URL is mapped to the university's domain system.
- ❖ **Backend Initialization:** The Node.js server is initialized. PM2 starts the application daemon, and the server establishes a persistent connection pool with the MongoDB Atlas cluster.
- ❖ **AI Engine Integration:** The NLP scripts are loaded into the server memory. A series of test vectors are processed to verify that the Cosine Similarity algorithm is functioning correctly within the production environment.

Stage 3: Post-Deployment Verification

- ❖ **Connectivity Testing:** A comprehensive test is conducted to verify that the API Gateway successfully routes requests for student login, internship posting, and profile forwarding.
- ❖ **Latency Monitoring:** The response time for the "Smart Matching" feature is measured to ensure that generating a match score takes less than 2 seconds under standard load.

Stage 4: Post-Deployment Maintenance and Monitoring

Once the system is live, continuous maintenance protocols are activated to ensure the longevity of the platform:

- ❖ **Automated Backups:** A schedule is set to perform daily automated snapshots of the Student Profile and Application collections, ensuring disaster recovery capabilities.

- ❖ **Real-Time Logging:** Tools like **Morgan** or **Winston** are used to log system activities. This allows the UIL Coordinator (Department Head) to audit login attempts, track successful placements, and identify any potential bottlenecks in the brokerage workflow.
- ❖ **Documentation:** Detailed user manuals are provided to the University Admin, explaining how to manage the server dashboard and interpret system logs.

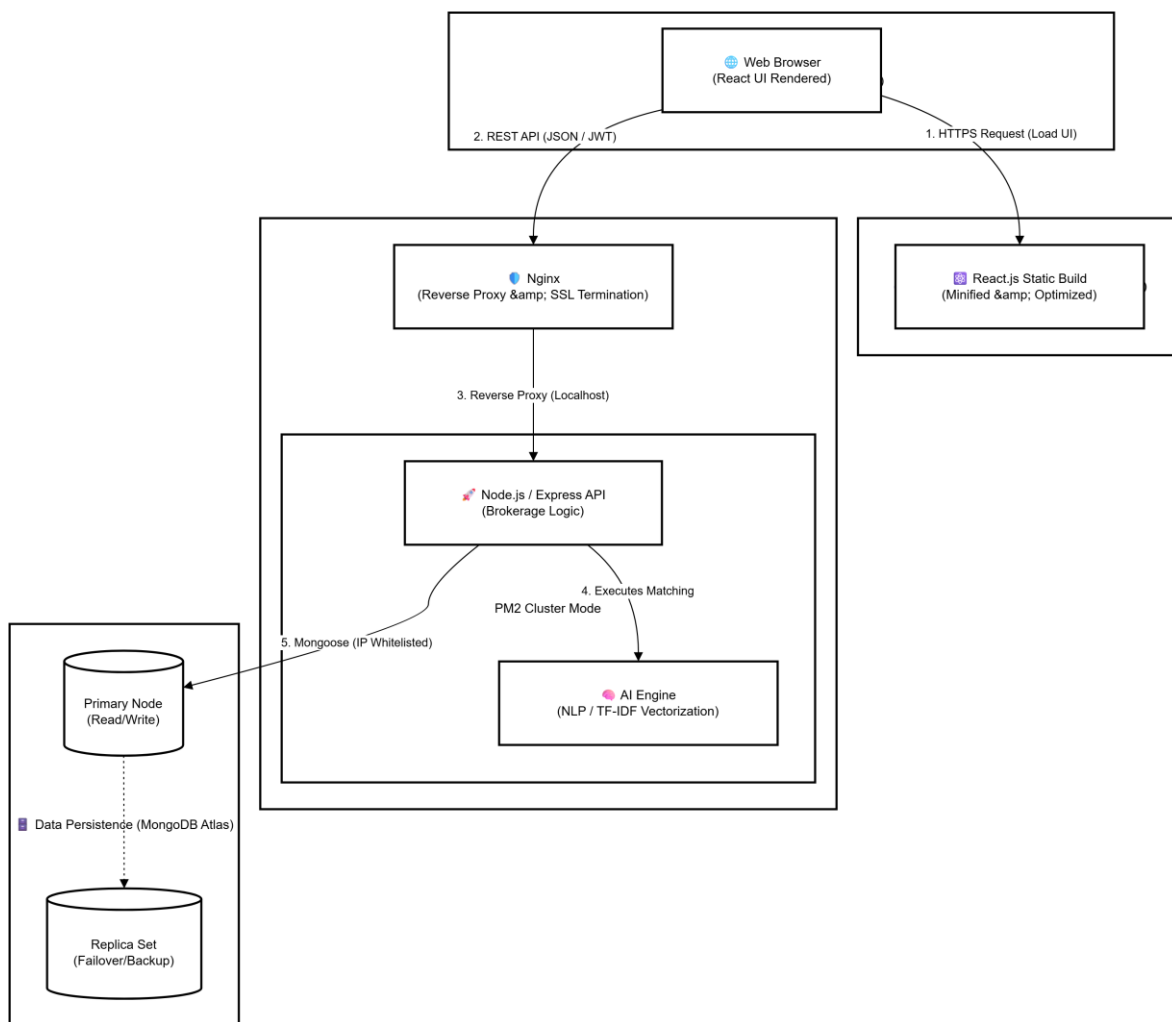


Figure 4-16: Deployment diagram

CHAPTER SIX

CONCLUSION AND RECOMMENDATION

6.1 CONCLUSION

The Smart Internship Placement for University Students project has successfully designed a web-based platform to modernize the internship linkage process at the University of Gondar. By transitioning from a manual, paper-based workflow to a digital "Managed Brokerage" system, this project addresses the critical challenges of inefficiency, data redundancy, and random student placement.

Key achievements of the system include:

- ❖ **AI-Driven Matching:** The implementation of the NLP Engine (using TF-IDF and Cosine Similarity) ensures that students are matched with industry opportunities based on mathematical compatibility between their skills and the organization's requirements, rather than random assignment.
- ❖ **Institutional Oversight:** Unlike open job boards, the system preserves the university's control through the UIL Coordinator module, allowing for the verification of industry partners and the auditing of student applications before forwarding.
- ❖ **Centralized Data Management:** Built on the MERN Stack, the system provides a persistent digital repository for student profiles and internship history, ensuring data integrity and facilitating long-term tracking.

In summary, this system bridges the gap between academia and industry by creating a secure, transparent, and intelligent environment that enhances the professional readiness of students.

6.2 RECOMMENDATIONS

Based on the current scope and limitations identified during the development of the project, the following recommendations are proposed for future enhancements:

- ❖ **Integration with University Registrar:** Currently, students manually input their grades. Future iterations should integrate directly with the University's Registrar database to automatically fetch and validate student GPAs and academic status to prevent fraud.
- ❖ **Expansion to Other Departments:** The current AI matching logic is optimized for the Department of Computer Science. The system should be scaled and the NLP

dataset expanded to support other faculties such as Engineering, Medicine, and Social Sciences.

- ❖ **Mobile Application Development:** Since the current system is a web-based application, developing native mobile apps (Android/iOS) would allow for realtime push notifications and GPS-based attendance tracking for students at internship sites.
- ❖ **Multi-Language Support:** To increase accessibility for local industry partners who may post requirements in local languages, the NLP engine should be upgraded to support Amharic alongside English.
- ❖ **Offline Functionality:** Implementing offline synchronization capabilities would ensure that UIL coordinators and students can access critical data even during internet outages on campus.

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